ECG-BASED ARRHYTHMIA DETECTION USING DEEP LEARNING: AN INTER-PATIENT COMPARISON WITH EXPLAINABLE AI AND EDGE DEPLOYMENT

A Thesis submitted in partial fulfillment for the requirement of the degree of

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by

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Heaven's Light is Our Guide

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CERTIFICATE

This is to certify that the thesis entitled **"ECG-Based Arrhythmia Detection Using Deep Learning: An Inter-Patient Comparison with Explainable AI and Edge Deployment"** by MD. Abu Huraira Mridha (Roll No. 2001120) and MD Rasheduzzaman Noor (Roll No. 2001142), has been carried out under my supervision. To the best of my knowledge, this thesis is an original one and has not been submitted anywhere for any Degree or Diploma.

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Abstract

Cardiovascular diseases are among the leading causes of death worldwide, and timely detection of cardiac arrhythmias from the electrocardiogram (ECG) is essential to prevent life-threatening complications. Manual ECG interpretation is time-consuming, requires expert cardiologists, and is vulnerable to inter-observer variability, which motivates automated computer-aided classification. This thesis develops and compares two deep-learning architectures for ECG beat classification: a hybrid CNN–BiLSTM model with temporal additive attention, and a lightweight multi-scale feature-augmented network (MSFT-Net) with squeeze-and-excitation gating and FiLM conditioning. Both models classify the five ANSI/AAMI EC57 beat classes (N, S, V, F, Q) from one-second R-peak-centred windows combined with rhythm, morphological, and template-similarity features under a record-disjoint, inter-patient evaluation protocol, together with a permissive intra-patient six-category ECG comparison that additionally includes a CHF-derived clinical-condition category. Under the inter-patient protocol, MSFT-Net achieved 89.45% accuracy and 87.25% macro-F1 with 167,090 parameters, outperforming the larger recurrent hybrid (82.00% accuracy, 73.26% macro-F1), while the intra-patient protocol reached 98.59% accuracy. The INT8-quantised MSFT-Net was deployed on an STM32 microcontroller with 185.06 ms per-beat inference latency, demonstrating that accurate, explainable, inter-patient arrhythmia classification is feasible on resource-constrained wearable hardware.

Keywords: ECG, arrhythmia classification, deep learning, MSFT-Net, CNN–BiLSTM–attention, inter-patient evaluation, XAI, Grad-CAM, microcontroller deployment

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List of Abbreviations

Abbreviation	Full Form
AAMI	Association for the Advancement of Medical Instrumentation
BiLSTM	Bidirectional Long Short-Term Memory
BIDMC	Beth Israel Deaconess Medical Center
CEA	Complex Engineering Activity
CEP	Complex Engineering Problem
CHF	Congestive Heart Failure
CNN	Convolutional Neural Network
CWT	Continuous Wavelet Transform
ECG	Electrocardiogram
EEE	Electrical & Electronic Engineering
FiLM	Feature-wise Linear Modulation
Grad-CAM	Gradient-weighted Class Activation Mapping
INT8	8-bit Integer Quantization
LSTM	Long Short-Term Memory
MCU	Microcontroller Unit
MIT-BIH	Massachusetts Institute of Technology–Beth Israel Hospital
MSFT	Multi-Scale Feature-Augmented (Network)
NSR	Normal Sinus Rhythm
PO	Program Outcome
QRS	QRS Complex of the Electrocardiogram
RAM	Random Access Memory
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
RR	Interval between successive R-peaks
RUET	Rajshahi University of Engineering & Technology
SE	Squeeze-and-Excitation
SMOTE	Synthetic Minority Over-sampling Technique
SNR	Signal-to-Noise Ratio
STM32	STMicroelectronics 32-bit Microcontroller
TFLite	TensorFlow Lite

XAI	Explainable Artificial Intelligence

List of Symbols

Symbol	**Description**	**Unit**
N	Normal beat class	–
S	Supraventricular ectopic beat	–
V	Ventricular ectopic beat	–
F	Fusion beat class	–
Q	Unclassifiable (paced) beat class	–
f_s	Sampling frequency	Hz
F_1	Harmonic mean of precision and recall	–
$+P$	Positive predictivity	–
Se	Sensitivity (recall)	–
p	Statistical significance level	–
Δ	Change in a quantity	–

Chapter 1

Introduction

Cardiovascular diseases (CVDs) continue to be a leading cause of death around the world. Among them, cardiac arrhythmias, irregularities of the heart rate caused by abnormal electrical signals in the heart, can lead to serious complications such as stroke, heart failure, and sudden cardiac arrest if they are not detected in time. This thesis develops a deep learning based computer-aided system for automated ECG arrhythmia classification and evaluates it under a clinically meaningful, record-disjoint protocol that is suitable for wearable, resource-constrained deployment.

1.1 Overview

The electrocardiogram (ECG) is the most widely used non-invasive diagnostic tool for assessing cardiac electrophysiology and diagnosing arrhythmias. However, ECG reading is time consuming and is normally performed manually, which makes it vulnerable to inter-observer variability and limits its use for long-term or continuous monitoring. A single 24-hour ambulatory record can contain on the order of 10^5 heart beats, and annotating such a record manually is expensive and impractical at scale. There is therefore a clear and growing need for automated, high-throughput screening systems that can classify beats rapidly, precisely, and reliably.

Recent advances in deep learning have substantially improved automated ECG analysis. Convolutional neural networks (CNNs) extract ECG morphology directly from the raw signal, while recurrent networks such as long short-term memory (LSTM) architectures capture the temporal relationships between successive cardiac beats. Self-attention and Transformer encoders have also been introduced to focus the model on the most informative signal regions, increasing both classification performance and interpretability. Channel-recalibration mechanisms such as squeeze-and-excitation (SE) blocks and feature-wise conditioning schemes like FiLM further improve the ability of a network to discriminate morphologically similar classes at modest parameter cost. Finally, explainable artificial intelligence (XAI) techniques such as Grad-CAM and Integrated Gradients provide visual explanations of model decisions, which helps build clinician confidence in automated systems.

In this thesis, a hybrid CNN–BiLSTM–attention model and a lightweight multi-scale feature-augmented network (MSFT-Net) are designed, trained, and compared for five-class ECG beat classification under a strict record-disjoint inter-patient protocol, and the best model is quantised and deployed on an STM32 microcontroller.

1.2 Background and Motivation

Despite the progress described above, several key issues remain unresolved in the automated ECG classification literature. Many existing studies focus on maximising classification accuracy while providing little transparency into the decision-making process of the deep learning model. Public ECG datasets are skewed, noisy, and dominated by normal beats, which reduces the capability of models trained on them to recognise rare but clinically important abnormal classes.

An even more serious, and less frequently discussed, issue concerns the evaluation protocol. A large portion of the literature reports results under a *beat-level* split in which beats from the same patient appear in both training and test sets. Because consecutive beats of one patient are strongly correlated in morphology and rhythm, a sufficiently expressive model can simply memorise the patient-specific beat templates and recall them at test time. The resulting figures measure within-patient recall rather than generalisation to unseen patients, and they are not comparable to inter-patient results, which typically reach much lower performance levels, especially for minority classes. The ANSI/AAMI EC57 standard recommends the record-disjoint DS1/DS2 partition of de Chazal et al. [3] precisely to avoid this problem, yet recent systematic reviews of the field show that both the standard partition and proper class grouping are frequently bypassed [4, 5].

Finally, the five AAMI superclasses are not equally difficult, and each resistant class fails for different reasons. A supraventricular ectopic beat (S) is defined by *prematurity*, a near-normal QRS complex arriving too early, so a fixed window centred on the R-peak alone cannot separate inter-patient S beats from normal (N) beats without timing information. A fusion beat (F) is, by definition, a morphological mixture of a normal and a ventricular depolarisation, so its features are inherently ambiguous. The unclassifiable class (Q) consists almost entirely of beats from four paced records pointed to by EC57, so its statistics rest on very few patients. These class-specific challenges motivate the feature engineering and targeted treatments developed in this work.

The motivation of this thesis is therefore threefold: (i) to build a hybrid deep learning classifier that learns ECG morphology and rhythm together, so that timing-sensitive classes such as S are recognised correctly; (ii) to evaluate it honestly under a record-disjoint inter-patient protocol so that the reported numbers reflect genuine generalisation to unseen patients; and (iii) to make the model compact and quantisable so that it can run in real time on a microcontroller-class wearable device, which is the computing environment available in real ambulatory monitoring.

1.3 Literature Review

Automated ECG arrhythmia classification research has developed along several largely independent dimensions (architecture design, class-imbalance mitigation, edge deployability, explainability, and evaluation methodology) that are rarely optimised together within a single study. This review considers each axis in turn, focusing on studies published mainly between 2020 and 2026 together with the foundational earlier works that established the field's core methods, and closes by identifying the gap that motivates the present work.

1.3.1 CNN–recurrent hybrids and channel attention

Convolutional and recurrent architectures remain the predominant building blocks for ECG beat classification. Kiranyaz et al. [6] made one of the first contributions, showing that patient-specific 1-D convolutional classifiers can run in real time on wearable devices; the method is not scalable to population level, however, because the filters are tuned per person. Channel-recalibration mechanisms were later transplanted into ECG backbones: Zhang et al. [7] proposed SE-ECGNet, a multi-scale residual network whose SE gating "reweighs" convolutional channels and improves discrimination between morphologically similar beat types at negligible parameter cost. Zheng et al. [8] extended this idea with a two-branch, multi-input CNN that fuses short-window and long-window spectrograms of the same beat, reporting 99.13% and 95.84% accuracy, and 94.46% and 95.91% macro-F_1, on the MIT-BIH and SPH databases respectively. A recurring finding of this line of work is that narrowly directing attention at the channel level does not by itself close the performance gap between the majority normal class and the clinically important minority ectopic classes.

1.3.2 Residual, spatial, and self-attention architectures

A second stream of research places attention at the spatial level. Xu et al. [9] fused residual skip connections with a U-Net like attention gate (RA-CNN) and reported 98.5% accuracy on a strict inter-patient DS1/DS2 split, with class-wise F_1 scores of 82.8% and 91.7% for the supraventricular and ventricular classes, figures that are informative precisely because they are reported under the more difficult inter-patient regime. Full Transformer self-attention has also been applied: Islam et al. [10] proposed CAT-Net, a cascaded convolutional, attention, and Transformer network for single-lead ECG classification that reached 99.14% accuracy on MIT-BIH by learning long-range inter-beat dependencies that fixed-receptive-field CNNs cannot represent; the authors note, however, that the model is too large for wearable deployment. Xia et al. [11] reported a lighter design combining a CNN front end with a denoising autoencoder and a Transformer encoder, evaluated explicitly under the inter-patient paradigm, and Anjum et al. [12] proposed a temporal Transformer-based fusion framework for morphological (rather than rhythmic) arrhythmia classes. Taken together, these studies suggest that self-attention is not a substitute for, but rather a complement to, convolutional morphology extraction, a finding that directly motivates the dual-branch CNN–Transformer design considered in this thesis.

1.3.3 Hybrid CNN–BiLSTM–attention frameworks and gated fusion

Recurrent sequence modelling is frequently combined with convolutional and attention modules to capture morphology and rhythm simultaneously. Mechichi and Benzarti [13] proposed a CNN–CBAM–BiLSTM nested network in which convolutional block attention rectifies spatial and channel features and a BiLSTM layer models beat-to-beat dependencies, reporting 99.20% accuracy, 97.50% sensitivity, and 98.29% mean F_1 on MIT-BIH after SMOTE-based rebalancing; the authors explicitly identify the BiLSTM stage as the main computational burden, a limitation that recurs throughout the lightweight-deployment literature. The family of architectures also disagrees on how parallel branches should be integrated: concatenation, summation, and learned gating are all used, and the choice of fusion mechanism carries measurable penalties for minority-class recall rather than being a neutral implementation detail. Feature-wise linear modulation (FiLM) [14] offers a lightweight, per-channel conditioning alternative for injecting auxiliary information.

1.3.4 Image-based and time-frequency representations

For the three-class separation of arrhythmia (ARR), congestive heart failure (CHF), and normal sinus rhythm (NSR), converting the 1-D signal into a 2-D time-frequency image has been fruitful. Olanrewaju et al. [15] classified scalograms obtained with the continuous wavelet transform (CWT) using an AlexNet-based deep network, reaching 98.7% accuracy on the ARR/CHF/NSR corpus. Madan et al. [16] fused a 2-D CNN with an LSTM stage running over scalogram sequences to reach similar accuracy, while Prusty et al. [17] paired SIFT features with a CNN classifier to achieve 99.78% accuracy at the cost of an extra, non-differentiable feature-extraction stage that complicates end-to-end optimisation. These studies confirm the discriminative power of image-based representations for ARR/CHF/NSR, but none addresses microcontroller-class deployment or a record-disjoint inter-patient partition, a limitation that is revisited in the evaluation-critique part of this review.

1.3.5 Class imbalance and synthetic minority augmentation

The fusion (F) and supraventricular (S) classes together contain fewer than 5% of the beats in MIT-BIH, so class-imbalance treatment is near-ubiquitous in the literature. The most popular baseline is SMOTE-family oversampling [18], used directly in several of the works cited above; more recent generative approaches train conditional or Wasserstein GANs to synthesise physiologically plausible minority-class beats, achieving up to 4–9 percentage points of improvement in minority precision when the synthetic beats are quality-screened against real-beat templates. A major drawback of these methods is that unscreened synthetic beats can be trivially distinguished from real ones by a downstream discriminator; an almost-perfect discriminator AUC is reported as direct proof of an under-converged generator rather than of class-balancing success. This diagnostic (training a real-versus-synthetic discriminator and using its AUC as a quality-control gate) is adopted as a validation step in the augmentation pipeline of the present study.

1.3.6 Lightweight and edge-deployable architectures

A growing body of literature considers deployment directly on microcontroller-class hardware rather than treating quantisation as an afterthought. Busia et al. [19] designed a tiny Transformer with only 6k parameters that achieves 98.97% accuracy on five AAMI classes with sub-8-bit integer inference, executing one inference in 4.28 ms at 0.09 mJ on the ultra-low-power GAP9 processor, evidence that self-attention, previously thought to require too much memory for microcontrollers, can be made competitive with convolutional baselines. At the opposite end of the hardware spectrum, Zambrano-de la Torre et al. [20] implemented a quantised 1-D CNN on an 8-bit Arduino UNO (32 KB flash, 2 KB SRAM), achieving 97.6% accuracy on a reduced three-class problem with a footprint below 24 KB INT8. The authors acknowledge, however, that their evaluations use a beat-level split without patient separation, a known source of overly optimistic numbers. The three-way trade-off between representational capacity, efficiency, and feasible evaluation is the central concern of this literature, and it is what motivates the "dual-model" design (an accuracy-oriented hybrid and a deployment-oriented lightweight network) used in the present study.

1.3.7 Evaluation methodology: the inter-patient/intra-patient divide

Almost all of the studies above share a methodological thread that deserves separate treatment. Two recently published systematic reviews of the ECG arrhythmia classification literature, covering the years 2017–2024, found that a substantial fraction of published works either fail to state whether their split is inter-patient or intra-patient, or report only the more favourable intra-patient figure [4, 5]. Silva et al. [4] formalise an "Embedded, Clinical, and Comparative Criteria" (E3C) checklist and find that very few surveyed studies satisfy inter-patient partitioning, AAMI-compliant class grouping, and embedded feasibility simultaneously. This reporting inconsistency has real consequences: the 85–100% accuracy figures reported under beat-level splits collapse dramatically under a record-disjoint protocol (equivalent architectures have been shown to drop to below 60% inter-patient accuracy) because beats from the same patient are morphologically correlated and the model exploits this correlation to memorise patient-specific templates rather than to generalise. The headline accuracy values reported throughout the literature are therefore not directly comparable unless the evaluation protocol is controlled for.

1.3.8 Explainability

Clinical adoption of automated ECG classifiers depends on the decision process being transparent rather than a black box. Grad-CAM, originally designed for 2-D image classifiers, has been adapted to 1-D ECG backbones to produce beat-level saliency maps that can be checked against known P–QRS–T morphology [1]. Integrated Gradients offers a complementary, axiomatically based attribution that does not rely on the selection of a specific convolutional layer [2]. Grad-CAM remains the most common method in the XAI-for-ECG literature and is normally applied to verify qualitatively whether the classifier's attention falls on the QRS complex and T-wave rather than on dataset-specific artefacts. Quantitative validation of this alignment (for example by deletion curves or intersection-over-union against cardiologist-annotated intervals) is reported far less often, and this thesis treats it as a faithfulness challenge rather than assuming correctness from visual inspection alone.

1.3.9 Summary and identified gap

The surveyed literature documents separately that: (i) channel and spatial attention improve discrimination of morphologically similar beats [21, 7, 9, 8]; (ii) long-range dependencies that convolutional or recurrent layers alone cannot capture are effectively handled by transformer self-attention [10, 11, 12, 22]; (iii) gated, learnable fusion of parallel branches outperforms naive concatenation; (iv) quality-screened generative or SMOTE-based augmentation recovers minority-class recall [18]; (v) models with sub-10k parameters can achieve competitive accuracy with INT8 quantisation on microcontroller-class hardware [19, 20]; and (vi) none of the accuracy figures above is trustworthy unless it is reported under a protocol that withstands inter-patient scrutiny [4, 5, 3]. What no single study has combined is a clearly stated record-disjoint protocol, explicit and ablated treatment of both of the hardest AAMI classes (fusion and supraventricular), a faithfulness-tested explainability layer, and a measured, rather than asserted, microcontroller deployment of the full inter-patient model. Two architectural families (a CNN–BiLSTM–attention hybrid and a gated-fusion lightweight CNN) are therefore evaluated in this thesis under identical data splits so that they can be compared statistically rather than across incommensurable papers. This gap is the direct driver of the research described in the remainder of this book.

1.4 Research Gap

Existing automated ECG classification systems suffer from three interlocking weaknesses. First, most published results are obtained under beat-level or otherwise leaky splits that inflate accuracy by letting the model memorise patient-specific templates, so the reported numbers do not reflect how the system would behave on an unseen patient. Second, the two most clinically dangerous minority classes (fusion and supraventricular ectopic beats) remain poorly recognised because they require timing and template information that raw-waveform-only models do not receive. Third, most accurate models are too large to run on the microcontroller-class hardware found in wearable monitors, and the few lightweight models that do run on such hardware are typically validated under optimistic, non-standard protocols. Therefore, there is a need for a compact, feature-augmented deep learning classifier that is trained and tested under a strictly record-disjoint inter-patient protocol, treats the resistant S and F classes explicitly, verifies its explainability quantitatively, and is demonstrated to run in real time on an actual microcontroller.

1.5 Objective

The objectives of this thesis are:

1. To design a hybrid CNN–BiLSTM–attention model and a lightweight multi-scale feature-augmented network (MSFT-Net) that classify the five ANSI/AAMI EC57 beat classes (N, S, V, F, Q) from one-second R-peak-centred ECG windows combined with rhythm, morphological, and template-similarity auxiliary features.
2. To evaluate both architectures under a strictly record-disjoint, inter-patient DS1/DS2 protocol (and a permissive intra-patient six-category ECG protocol that adds a CHF-derived clinical-condition category for comparison), reporting per-class sensitivity and positive predictivity with statistical significance testing.
3. To analyse the contribution of each architectural and feature component through a controlled ablation study, and to verify the faithfulness of the Grad-CAM and Integrated-Gradients explanations using deletion curves.
4. To quantise the best inter-patient model to INT8 and deploy it on an STM32 microcontroller, measuring accuracy retention, model size, latency, and memory footprint.

1.6 Thesis/Project Outline

The remainder of this book is organised as follows.

Chapter 2 describes the methodology adopted in this work: the MIT-BIH and auxiliary databases, beat extraction and filtering, the record-disjoint partitioning scheme, the auxiliary feature vector, the two network architectures, the training and imbalance-handling procedures, and the statistical evaluation and explainability methods.

Chapter 3 presents the results obtained: the exploratory intra-patient six-category ECG study, the inter-patient five-class study with its AAMI EC57 reporting, the ablation and fusion-beat analyses, the explainability and robustness results, and the microcontroller deployment measurements.

Chapter 4 examines the social and environmental influence of the work, including its financial, health, safety, legal, and sustainability implications for Bangladesh.

Chapter 5 explains how the work addresses the program outcomes, complex engineering problems and complex engineering activities defined in the EEE 4000 manual.

Chapter 6 concludes the book and outlines the future plan.

The work was carried out over the fourth year of the B.Sc. in Electrical & Electronic Engineering programme at RUET.

Chapter 2

Methodology

This chapter describes the complete experimental pipeline of the thesis: the public databases used, the beat-extraction and preprocessing stages, the record-disjoint partitioning scheme, the auxiliary feature vector, the two network architectures (the CNN–BiLSTM–attention hybrid and MSFT-Net), the training and imbalance-handling procedures, and the statistical evaluation, explainability, and deployment methods.

2.1 Introduction

The aim of this chapter is to present the design of the study in a way that allows exact reproduction. Section 2.2 states the overall research design; Section 2.3 describes the databases and computing tools; Section 2.4 describes signal preprocessing, beat extraction, and the auxiliary features; Section 2.5 gives the mathematical formulation of the network architectures; Section 2.6 details the proposed pipeline; Section 2.7 describes the training workflow and imbalance handling; and Section 2.8 defines the performance evaluation criteria and statistical procedures.

2.2 Research Design

The work is a retrospective computational study: pre-recorded, publicly available, and de-identified ECG data are used to develop and test beat classifiers. The main analysis is a supervised five-class classification problem (N, S, V, F, Q) solved with two deep learning architectures, with a secondary six-category ECG experiment used to quantify the effect of the evaluation protocol. A record-disjoint, inter-patient partitioning scheme is the central design constraint: no patient contributes beats to more than one partition, so that the reported performance measures genuine generalisation to unseen patients. Secondary design constraints are (i) class-balancing operations are applied to the training partition only, (ii) all auxiliary statistics are computed from training beats only, and (iii) the final model must fit within the flash, RAM, and latency budgets of a microcontroller-class wearable device.

2.3 Materials & Equipment

2.3.1 Databases

The primary data source is the MIT-BIH Arrhythmia Database, which comprises 48 half-hour, two-channel ambulatory electrocardiograms sampled at 360 Hz with beat annotations [23, 24]. Two auxiliary databases extend the corpus: the MIT-BIH Normal Sinus Rhythm Database (18 recordings, used to augment the normal class) and the BIDMC Congestive Heart Failure Database (15 recordings of about 20 hours each, sampled at 250 Hz) [24, 25]. Beat-level labels originate only from the MIT-BIH Arrhythmia annotations, which are grouped into the five ANSI/AAMI EC57 classes. The NSR recordings carry no beat annotations and are folded into the normal class; the CHF recordings likewise carry no beat annotations and are labelled purely by database provenance as a separate CHF-derived clinical-condition category in the secondary experiment. CHF is a cardiovascular condition rather than an AAMI beat class, so the secondary task is a six-category ECG problem, not a six-class arrhythmia problem. All data are de-identified and publicly available through PhysioNet, so no ethical approval was required for this retrospective study.

2.3.2 Software and hardware tools

The models were implemented in Python 3 using TensorFlow (with Keras), NumPy, and scikit-learn for metric computation. Training was performed on a GPU-equipped workstation. Deployment measurements were carried out on an STM32 microcontroller (128 KB RAM, 512 KB flash) using the TensorFlow Lite Micro runtime. Random seeds in Python, NumPy, and TensorFlow were set to 42 for reproducibility.

2.4 Data Analysis Techniques

2.4.1 Signal preprocessing and beat extraction

Each ECG signal was filtered with a second-order Butterworth band-pass filter (cut-off frequencies 0.5 and 40 Hz) applied as a zero-phase filter. For the MIT-BIH Arrhythmia Database, the reference beat annotations were used; where no beat label was provided, R-wave peaks were detected with the Pan–Tompkins algorithm [26], which combines differentiation, squaring, moving-window integration, and adaptive thresholding. Signals sampled at frequencies other than 360 Hz were linearly resampled to 360 Hz. For each detected R-peak, a one-second window (360 samples) centred on the peak was extracted; windows extending beyond the recording boundary were omitted, and the MLII lead was used whenever available. Scaling parameters for each retained waveform were estimated using only the training partition. The possible impact of sampling and bandwidth differences across sources was tested separately by resampling all beats to a common 128 Hz.

2.4.2 Auxiliary feature extraction

Four signal windows of raw waveform were used together with an auxiliary feature vector of 107 dimensions. Rhythm features included pre- and post-RR intervals and their ratio, RR intervals normalised relative to rolling local and record-level medians, the local RR coefficient of variation, and indices of prematurity and compensatory pause. These quantities were computed from the entire R-peak sequence before beat filtering, then indexed to the retained beats. Morphological features were computed over the interval 250–80 ms prior to the R-peak, including P-wave morphology and QRS duration and shape features, and were compared against high-quality normal-beat templates from the same record as well as global normal and ventricular templates. Only raw training beats were used to construct the global templates and the synthetic normal-to-ventricular fusion continuum. The auxiliary features were scaled using training-partition statistics (median and interquartile range).

2.5 Mathematical Formulation

2.5.1 CNN–BiLSTM–attention hybrid

The hybrid model, whose block diagram is shown in Figure 2.1, processes the waveform with three cascaded 1-D CNN blocks with 64, 128, and 256 filters, each followed by batch normalisation, ReLU activation, and max pooling. The sequence output of the final convolutional stage is passed to a bidirectional LSTM with 128 units per direction, and the LSTM output is pooled with additive temporal attention:

$$\alpha_t = \frac{\exp\left(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_t)\right)}{\sum_{t'} \exp\left(\mathbf{v}^\top \tanh(\mathbf{W}\mathbf{h}_{t'})\right)} \tag{2.1}$$

where $\mathbf{h}_t$ is the hidden state at time step t, and $\mathbf{W}$ and $\mathbf{v}$ are learnable attention parameters. The pooled waveform representation $\sum_t \alpha_t \mathbf{h}_t$ is combined with a 48-unit dense branch that processes the auxiliary feature vector, and the concatenation feeds the dense softmax classifier.

2.5.2 MSFT-Net

The comparison model, MSFT-Net, begins with three parallel depthwise-separable convolutional blocks with kernel sizes 5, 11, and 21 samples followed by a single pointwise fusion layer. Three residual, depthwise-separable stages follow, each including squeeze-and-excitation (SE) channel gating and feature-wise linear modulation (FiLM) conditioned on the auxiliary vector. SE gating reweights the channels of a feature map:

$$\mathbf{x}' = \sigma\left(\mathbf{W}_2\, \mathrm{ReLU}(\mathbf{W}_1\, \mathrm{GAP}(\mathbf{x}))\right) \odot \mathbf{x} \tag{2.2}$$

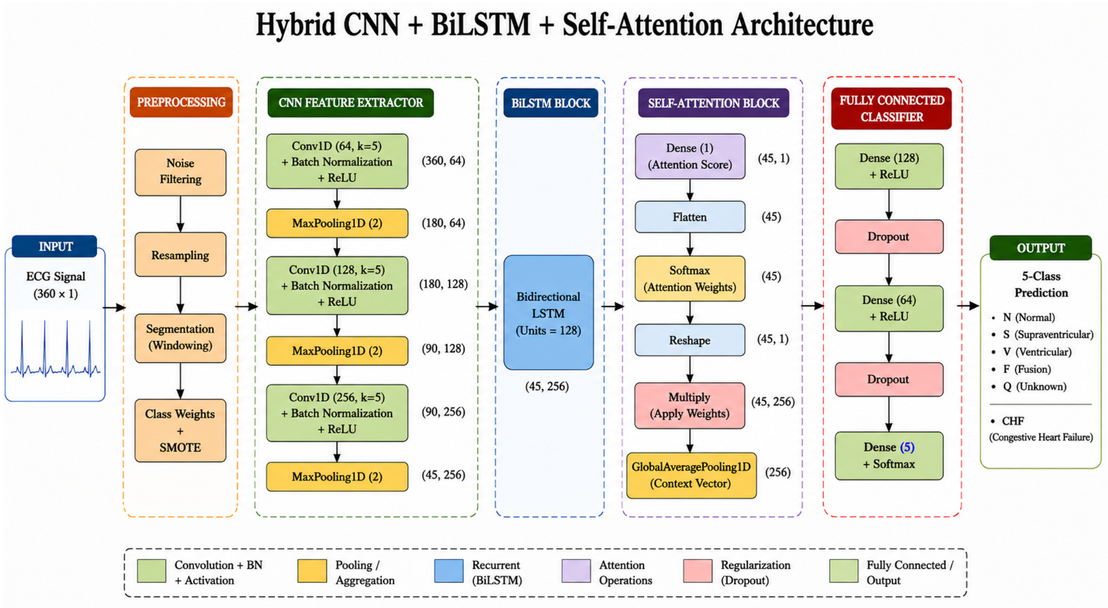

Figure 2.1. Block diagram of the intra-patient CNN–BiLSTM–attention hybrid architecture

where GAP is global average pooling, σ is the sigmoid function, and $\odot$ is the Hadamard product. FiLM applies per-channel affine conditioning derived from the auxiliary vector:

$$\mathbf{y} = \gamma(\mathbf{a}) \odot \mathbf{x} + \beta(\mathbf{a}) \tag{2.3}$$

where γ and β are generated by dense layers from the auxiliary vector $\mathbf{a}$. The overall data flow of MSFT-Net – the inter-patient deployment model – is summarised in Figure 2.2. The same late auxiliary-feature fusion, global average pooling, and classification head are used as in the hybrid. Two additional CNN-only and CNN–BiLSTM models without attention were trained as ablation baselines, and the microcontroller implementation used a separate single-input CNN.

2.6 Proposed Methodology

2.6.1 Sampling, labels, and record-disjoint partitioning

The five AAMI groups were mapped to the MIT-BIH annotation symbols. Paced records 102, 104, 107, and 217 were excluded from the N/S/V/F analysis; because this rule filters out most Q beats, beats from these records were kept when the Q-class policy was enabled and excluded otherwise. Results for Q are therefore interpreted with caution, as they rest on very few patients.

Records, rather than individual beats, were assigned to partitions to avoid patient leakage, following the de Chazal convention of separating the records used for model development from those used for testing [3]. A validation subset of the development records, sharing no records with the test set, was reserved for early stopping. Normal-sinus-rhythm and congestive-heart-failure records used in the auxiliary analyses were likewise assigned by record identifier. Programmatic checks ensured that no record appeared in more than one partition and that all required classes were present in the training and test partitions.

Instead of enforcing a single global beat cap, beats were sampled with per-record quotas via evenly spaced indices, so that every record contributed observations and long recordings did not dominate the corpus. Where possible, rare S and F beats were preserved. Splitting, resampling, augmenting, scaling, template building, and imbalance correction were applied only to the training partition.

2.6.2 Fusion-beat treatment

A staged intervention was designed for the fusion class: (i) global-average pooling in place of attention pooling, (ii) F-specific augmentation, (iii) F/V hard-negative mining, and (iv) auxiliary F-vs-N and F-vs-V heads. Each stage was studied in isolation in a dedicated ablation arm so that its contribution could be measured rather than guessed at.

2.6.3 Microcontroller model

A separate, single-input CNN was derived for the microcontroller implementation. The model was quantised to full INT8 with post-training quantisation using representative calibration beats, checked against the TensorFlow Lite Micro operator set, and evaluated for accuracy loss, latency, and memory footprint.

2.7 System Workflow

2.7.1 Training and imbalance handling

Models were trained with the Adam optimiser, categorical cross-entropy loss, a batch size of 64, and up to 50 epochs. Early stopping, learning-rate decay on plateau, and checkpointing of the best model were all applied using the validation set. Three class-imbalance strategies were compared: no correction, a class-weighted loss function, and SMOTE oversampling applied to the training data only, not the test data [18]. Class weights were calculated after oversampling and were never combined with SMOTE. Auxiliary experiments applied conservative class-specific augmentation, hard-negative mining (for F-vs-V and Q), and auxiliary heads (for F-vs-N, F-vs-V, and Q-vs-rest discrimination).

2.8 Performance Evaluation Criteria

2.8.1 Primary metrics

Class-specific sensitivity (Se) and positive predictivity ($+P$), reported as both gross (pooled-beat) and average (per-record) figures, were the primary results, as recommended by the ANSI/AAMI EC57 standard [27]. Precision–recall curves, precision at fixed recall, macro-F_1, and accuracy were also reported. The macro-F_1 is defined as the unweighted mean of the per-class F_1 scores, where each $F_1 = 2 \cdot P \cdot R/(P + R)$ with P the precision and R the recall.

2.8.2 Statistical analysis

95% confidence intervals for proportions were constructed using Wilson's method [28]. The 95% confidence interval for macro-F_1 was estimated with a nonparametric bootstrap using 2,000 resamples. Paired test predictions were compared with the continuity-corrected McNemar test [29], adjusted for multiple comparisons with the Holm step-down procedure. All tests were two-tailed, and $P < 0.05$ was considered statistically significant.

2.8.3 Robustness and provenance

Robustness was assessed under additive Gaussian noise (signal-to-noise ratio from 20 dB down to 0 dB) and under increasing simulated baseline wander. Source confounding was probed with a logistic-regression discriminator trained to distinguish normal beats of the MIT-BIH database from those of the normal-sinus-rhythm database, followed by a re-analysis of the bandwidth-equalisation experiment.

2.8.4 Explainability

Model explanations were produced with Grad-CAM [1], attention weights, and Integrated Gradients [2]. Faithfulness was assessed with deletion curves, which plot the decrease in target-class probability after successively masking the most highly attributed samples against the decrease from random masking.

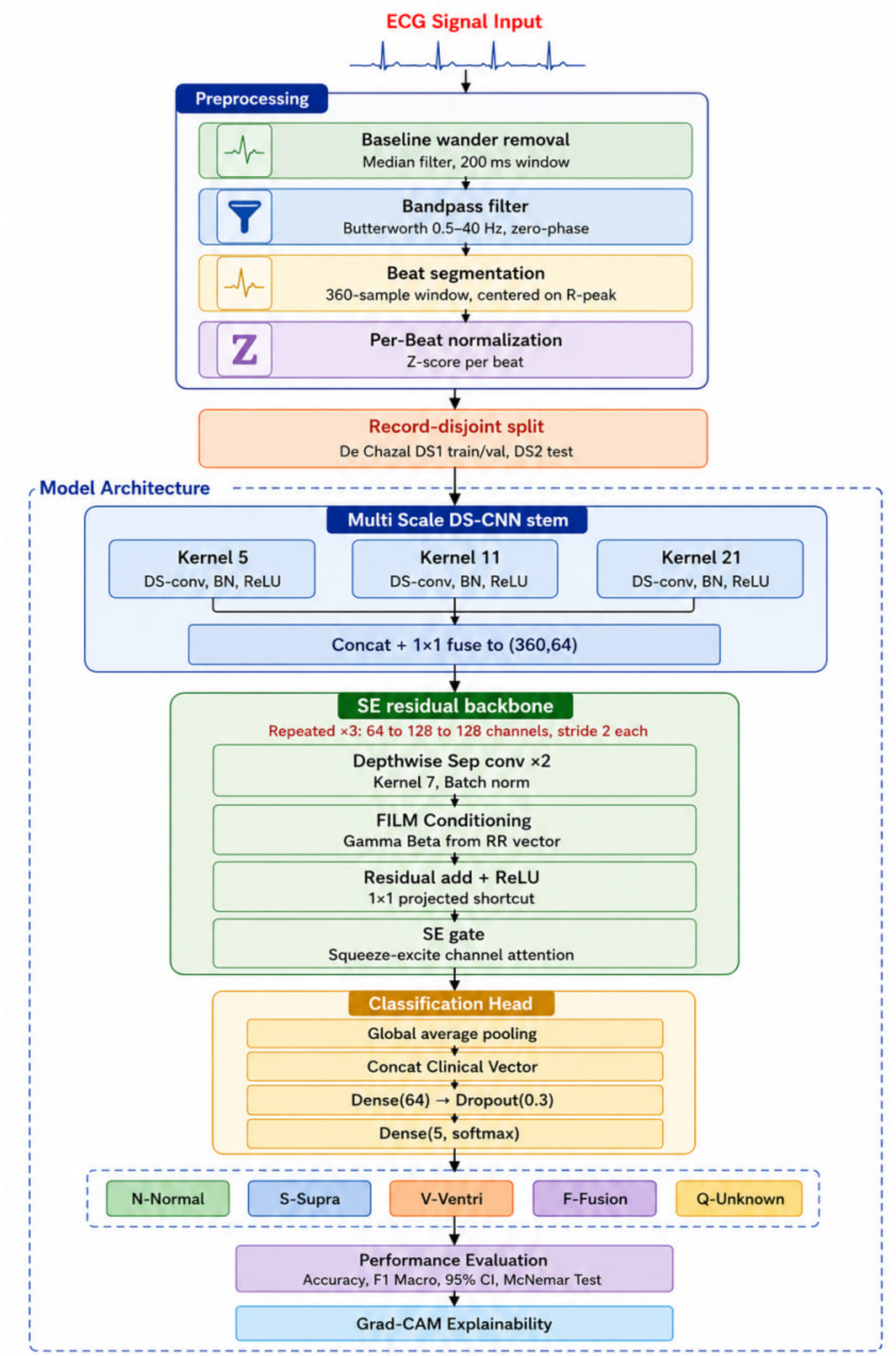

Figure 2.2. Block diagram of the inter-patient MSFT-Net architecture with multi-scale stem, squeeze-and-excitation, and FiLM conditioning

Chapter 3

Results and Discussion

This chapter presents the results of the two evaluation protocols used in this thesis and discusses their meaning. Section 3.1 introduces the two protocols and their best-model results; Section 3.2 reports the permissive intra-patient six-category ECG study; Section 3.3 reports the record-disjoint inter-patient five-class study, including the AAMI EC57 statistics, ablation analyses, explainability, robustness, and microcontroller deployment; Section 3.4 synthesises the two protocols; and Section 3.5 closes with a summary of the findings.

3.1 Introduction

The proposed pipeline is evaluated under two deliberately different protocols, and the contrast between them is itself one of the main themes of this investigation. The first is an *inter-patient*, record-disjoint protocol based on the five ANSI/AAMI EC57 classes: normal (N), supraventricular ectopic (S), ventricular ectopic (V), fusion (F), and unknown/paced (Q) [27], in which no patient contributes beats to more than one partition. This is the regime under which arrhythmia classifiers are comparable across studies, and it is much more difficult because the model never sees a test patient during training. The second is an *intra-patient* protocol with a stratified train/validation/test split and SMOTE balancing [18], extended with a sixth category comprising ECG segments derived from the BIDMC Congestive Heart Failure Database (CHF). Because CHF is a cardiovascular clinical condition rather than an ANSI/AAMI EC57 beat class, this arm is described throughout as a six-*category* ECG experiment rather than a six-class arrhythmia problem; it is the more permissive, within-database regime frequently found in the literature. The inter-patient protocol is the more stringent: it uses a record-disjoint split so that no test patient's beats appear during training, and it reports only the five AAMI EC57 classes. The intra-patient protocol, by contrast, allows the same patient's beats in both training and test partitions and adds a sixth, CHF-derived clinical-condition category. The accuracy gap between the two arms—0.8945 versus 0.9859—quantifies how much of the reported performance in the literature is an artefact of the assessment regime rather than genuine generalisation. Table 3.1 compares the two protocols and their best-model results. The intra-patient numbers cannot be compared with the inter-patient numbers; the inter-patient protocol is the clinically meaningful one because under a record-disjoint split none of the test patients' beats influenced training.

Table 3.1. Comparison of the inter-patient (Part A) and intra-patient (Part B) evaluation protocols and their best-model results

	Inter-patient (Part A)	**Intra-patient (Part B)**
Split	record-disjoint	stratified + SMOTE
Label set	5 AAMI classes (N/S/V/F/Q)	5 + CHF-derived category
Balancing on test	none	none
Test beats	9,112	4,329
Best accuracy	0.8945 (MSFT-Net)	0.9859
Best macro-F_1	0.8725 (MSFT-Net)	0.9684

3.2 Intra-Patient Six-Category Study

The second protocol relaxes the record-disjoint constraint: a stratified train/validation/test split is used, and a sixth category, comprising R-peak-centred ECG segments derived from the BIDMC Congestive Heart Failure Database, is added to the five AAMI beat categories to give a six-category problem (N/S/V/F/Q plus a CHF-derived clinical-condition category) evaluated on 4,329 test beats. This permissive, within-database regime is widely used in the literature, and it is reported here both for completeness and because its juxtaposition with the inter-patient results shows how much apparent performance is a by-product of the assessment regime.

3.2.1 Dataset Composition and Balancing

Beats were taken from the MIT-BIH Arrhythmia Database (48 records), the MIT-BIH Normal Sinus Rhythm Database (18 records), which was mapped to the normal class, and the BIDMC Congestive Heart Failure Database (15 records at 250 Hz) [23, 24, 25]. The normal and CHF contributions were each capped at 5,000 beats. Table 3.2 shows the composition from the raw extraction through the normal-class cap to the SMOTE-balanced training set, and Figure 3.1 displays the raw imbalance.

The data in Table 3.2 make two design decisions explicit. Capping the normal class prevents the majority class from dominating the SMOTE targets, and equalising the six training categories to 5,627 beats avoids class imbalance as a confounding factor during optimisation. Both choices are sensible for maximising within-database accuracy, but neither concerns generalisation between patients, precisely because the split is not record-disjoint, which is why the numbers below are substantially higher than those of the inter-patient part.

3.2.2 Training Dynamics and Overall Accuracy

The CNN–BiLSTM–Attention model was trained for up to 50 epochs with early stopping; Figure 3.2 shows how the learning curves evolve. The overall test accuracy of the full model

Table 3.2. Composition of the intra-patient six-category ECG corpus—five AAMI beat categories plus one CHF-derived clinical-condition category—from raw extraction through normal-class capping to the SMOTE-balanced training set

Class	Raw beats	After N-cap	Train (post-SMOTE)
N-Normal	10,000	5,000	5,627
S-Supra	2,781	2,781	5,627
V-Ventri	7,235	7,235	5,627
F-Fusion	802	802	5,627
Q-Unknown	8,038	8,038	5,627
CHF	5,000	5,000	5,627
Total	33,856	28,856	33,762

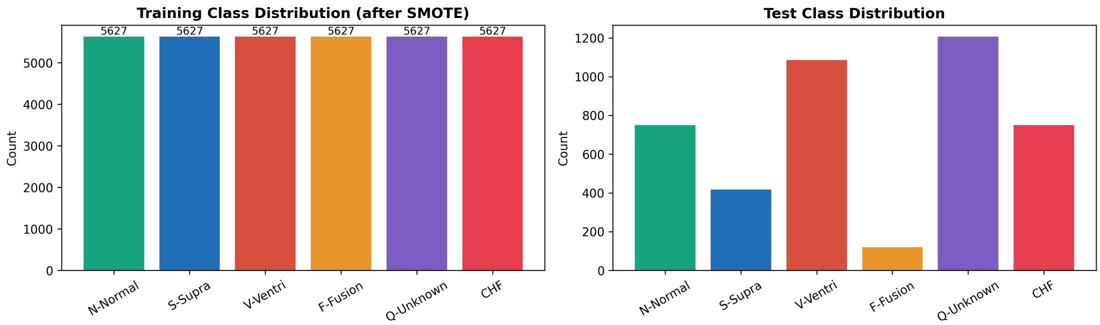

Figure 3.1. Raw class distribution of the six-category intra-patient corpus before balancing

is 0.9859, with a macro-F_1 of 0.9684. As Table 3.3 shows, five of the six categories achieve F_1 scores of 0.96 or better; only the fusion class remains difficult, with an F_1 of 0.84.

The CHF result must be interpreted with caution. As discussed in Section 3.3.10, the CHF segments come from a different database with a different sampling frequency than the arrhythmia records, so a large fraction of the model's CHF separability is likely due to a database “fingerprint” rather than cardiac morphology; it should not be read as evidence of a clinically generalisable CHF signature. It is for this reason that CHF was excluded from the rigorous five-class study. The one class that remains difficult even under the permissive protocol is fusion: its F_1 of 0.84 even when training and test patients coincide is evidence that the fusion difficulty is genuinely morphological, not an artefact of the record-disjoint split, and it retrospectively validates the dedicated fusion-handling pipeline of Section 3.3.6.

3.2.3 Confusion Structure and ROC

Figures 3.3 and 3.4 present the six-category confusion matrix and the one-versus-rest ROC curves. The residual error is concentrated in the fusion row and column, mirroring the inter-patient finding that fusion is the binding constraint on performance. The ROC curves are practically saturated, which is the logical result of a subject-mixed split: at test time the

Figure 3.2. Training and validation curves for the six-category hybrid under the intra-patient protocol

Table 3.3. Per-class precision, recall, and F_1 on the intra-patient six-category test set (representative single-run report); CHF denotes the CHF-derived clinical-condition category

Class	Precision	Recall	F_1	Support
N-Normal	0.99	1.00	0.99	750
S-Supra	0.95	0.97	0.96	417
V-Ventri	0.98	0.97	0.97	1,086
F-Fusion	0.83	0.86	0.84	120
Q-Unknown	1.00	0.99	0.99	1,206
CHF	1.00	0.99	0.99	750
Macro avg	0.96	0.96	0.96	4,329
Accuracy	0.9859			4,329

model has already seen the typical waveform of each patient, so detecting that patient's beat types is a much simpler problem than picking out beat types on a patient it has never seen. This is the quantitative centre of the cross-protocol comparison in Section 3.4.

3.2.4 Ablation: Contribution of the Recurrent and Attention Stages

Table 3.4 and Figure 3.5 isolate the contribution of the BiLSTM and attention stages by progressively building the architecture from a CNN-only baseline.

Table 3.4 contains an instructive result that directly contradicts what will be found under the record-disjoint split: the BiLSTM stage contributes a measurable accuracy gain of 0.0044 (and a macro-F_1 gain of 0.0056) under the subject-mixed split, while the attention stage adds nothing further (accuracy in fact decreases by 0.0009). The recurrent stage exploits patient-dependent temporal regularities that are present in a subject-mixed split but absent, by design, from the held-out patients of a record-disjoint split. In other words, part of what the recurrent stage learns is not transferable arrhythmia morphology but patient identity. This is a concrete, data-based demonstration of why the evaluation protocol changes not only the numbers reported but also the architectural conclusions drawn from them.

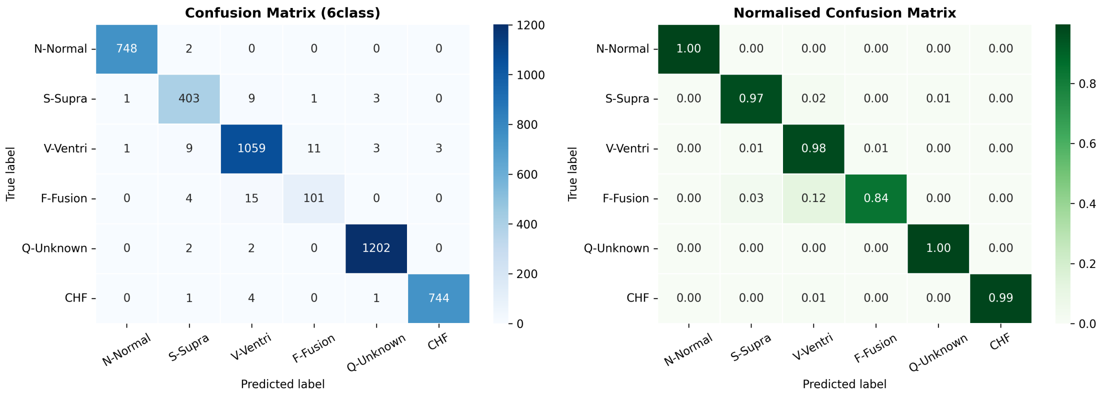

Figure 3.3. Confusion matrix for the six-category intra-patient ECG classification experiment

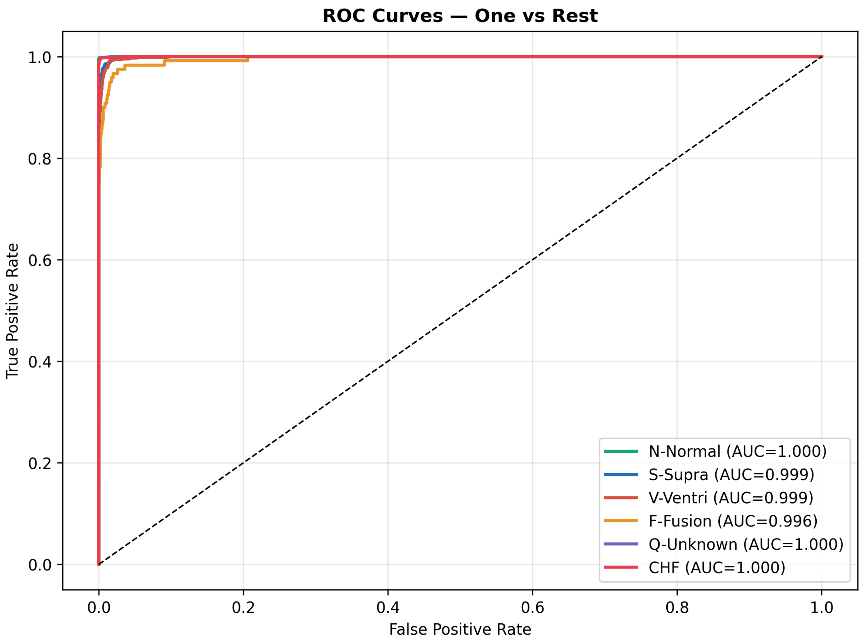

Figure 3.4. One-versus-rest ROC curves for the six-category intra-patient ECG classification experiment

Table 3.4. Architecture ablation under the intra-patient protocol

Model	Accuracy	Macro-F_1	ΔAcc
CNN only	0.9799	0.9604	–
CNN + BiLSTM	0.9843	0.9660	+0.0044
CNN + BiLSTM + Attention (proposed)	0.9834	0.9649	+0.0035

3.2.5 Model Compression

The six-category hybrid was also compressed for edge deployment. Dynamic-range quantisation reduced the converted lightweight model from 1.21 MB to 0.14 MB, an 88.42% reduction, but the measured test accuracy dropped to 0.1737, just above the 0.167 chance level

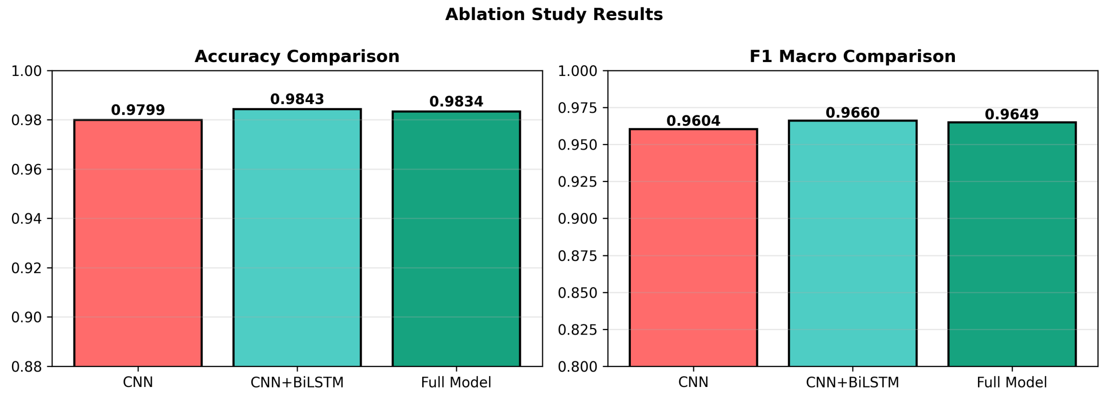

Figure 3.5. Accuracy and macro-F_1 across the three architecture arms under the intra-patient protocol

for the six-way problem. The dynamic-range conversion, applied without a representative calibration data set, did not maintain the decision function. This is reported as an intentional negative result: it contrasts with the inter-patient INT8 model of Section 3.3.11, which was quantised with proper post-training integer calibration. The practical lesson is that aggressive compression is feasible only when faithful low-bit quantisation is performed with appropriate calibration data, not by dynamic-range conversion alone.

3.2.6 Comparison with Reported Methods

Table 3.5 restates the comparison found in the executed notebook, comparing the intra-patient six-category model against representative reported ECG-classification methods. Because the studies are based on different databases, class groupings, and, most importantly, different train/test protocols, the comparison is indicative only; the numbers are not to be read as a like-for-like ranking.

Table 3.5 puts the intra-patient model in perspective: it achieves roughly the same performance as recent deep learning approaches, with two characteristics that most of them lack: a six-category label set that adds a CHF-derived category, and multi-database training. The most honest interpretation of this table, however, is comparative rather than competitive: several listed methods report slightly higher accuracy on the narrower five-class MIT-BIH problem, and none of the external numbers was obtained under a record-disjoint protocol. The results of Section 3.3 show that Part A is the more challenging and clinically important test, so the meaning of the present work is not to sit at the head of this table but to report both protocols transparently so that apparent performance differences can be attributed to their real source.

Table 3.5. Indicative comparison of the intra-patient six-category model with reported ECG-classification methods

Method	**Dataset**	**Categories**	**Accuracy**	**Macro-F_1**
This work (CNN+BiLSTM+Att. +XAI)	MIT-BIH+NSR+CHF	6	0.9859	0.9684
CNN–BiLSTM+Att	MIT-BIH	5	0.9920	0.9750
CNN+LSTM+Att	MIT-BIH+PTB	5	0.9930	0.9600
Transformer ECG	MIT-BIH	5	0.9832	0.9501
DenseNet1D	MIT-BIH	5	0.9756	0.9234
LSTM–FCN	MIT-BIH	5	0.9634	0.8967
SVM baseline	MIT-BIH	5	0.8976	0.7832

3.3 Inter-Patient (Record-Disjoint) Five-Class Study

All metrics in this section are calculated on a frozen inter-patient test partition of 9,112 beats from 26 patients of the MIT-BIH Arrhythmia Database [23, 24].

3.3.1 Dataset Composition and Partitioning

The beat corpus was assembled from three source types: 48 MIT-BIH records covering multiple rhythms, 18 MIT-BIH Normal Sinus Rhythm records used to augment the normal class, and a congestive-heart-failure set that was not included in the final label space following the provenance controls of Section 3.3.10. To keep the study EC57-conformant, the four paced records (102, 104, 107, 217) contribute Q beats only, so that the N/S/V/F supports are equivalent to a paced-free run. Table 3.6 summarises the class-wise beat counts, the record-disjoint split, and the record coverage; the resulting class imbalance is visualised in Figure 3.6. Two consequences of the sampling rules deserve emphasis. First, because the per-record quotas of Section 2.6 preserve every available S and F beat, the S and F totals equal their complete raw populations (2,781 and 802 beats, identical to the raw extraction of Section 3.2.1), whereas the abundant N, V, and Q classes are bounded by the quotas. The comparatively small V total (667 beats) therefore reflects quota sampling rather than the underlying prevalence of ventricular ectopy, and the per-class test supports should be read in this light.

The class distribution in Figure 3.6 is strongly imbalanced and clinically representative: fusion and ventricular beats occur rarely, yet they are the most expensive to miss when diagnosing. Training-only augmentation was applied to the minority classes (one perturbed copy of every training S beat and three additional copies of every training F beat), adding 1,971 beats and expanding the training set from 9,919 to 11,890 beats, while validation and test partitions were left untouched. The RR timing, QRS morphology, wavelet descriptors, and

Table 3.6. Beat-level composition of the five AAMI classes and the record-disjoint partitioning

Class	Records	Total beats	Train	Val.	Test
N-Normal	58	15,306	7,455	1,804	6,047
S-Supra	32	2,781	774	170	1,837
V-Ventri	25	667	283	51	333
F-Fusion	17	802	399	15	388
Q-Unknown	9	2,015	1,008	500	507
Total	–	21,571	9,919	2,540	9,112

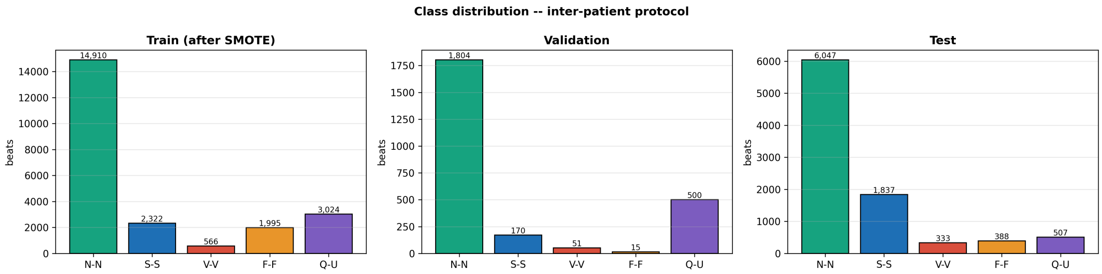

Figure 3.6. Class distribution of the assembled beat corpus under the inter-patient protocol

N/V-template-similarity features were computed from training statistics only, without using validation or test labels, so the scale of any feature or template cannot be contaminated by test information.

3.3.2 Training Dynamics

Figures 3.7 and 3.8 show the learning curves of the two architectures: the CNN–BiLSTM–Attention hybrid (572,358 parameters) and the lighter MSFT-Net (167,090 parameters) with multi-scale convolutions, squeeze-and-excitation, and FiLM conditioning.

The hybrid fuses quickly to the training objective: validation accuracy already exceeds 0.98 in the early epochs while macro-F_1 oscillates in the 0.71–0.77 band, and after early stopping at epoch 14 the epoch-2 weights are recovered. This is the expected behaviour of a high-capacity model: the capacity is spent recalling patient-specific local waveform details that do not transfer, and the learning-rate reductions of `ReduceLROnPlateau` (at epochs 7 and 12) purchase stability but not generalisation.

MSFT-Net tells a different story. Its validation curve is higher and less erratic because of its multi-scale stem and the auxiliary-vector dropout that breaks up the reliance on convolutions; the patient-specific FiLM conditioning visibly reduces validation variance. The two curves flank the central, experimentally found architectural conclusion of this part: for inter-patient generalisation, a compact convolutional model conditioned on auxiliary features is much larger in benefit than a much larger recurrent one, on exactly the metric that matters.

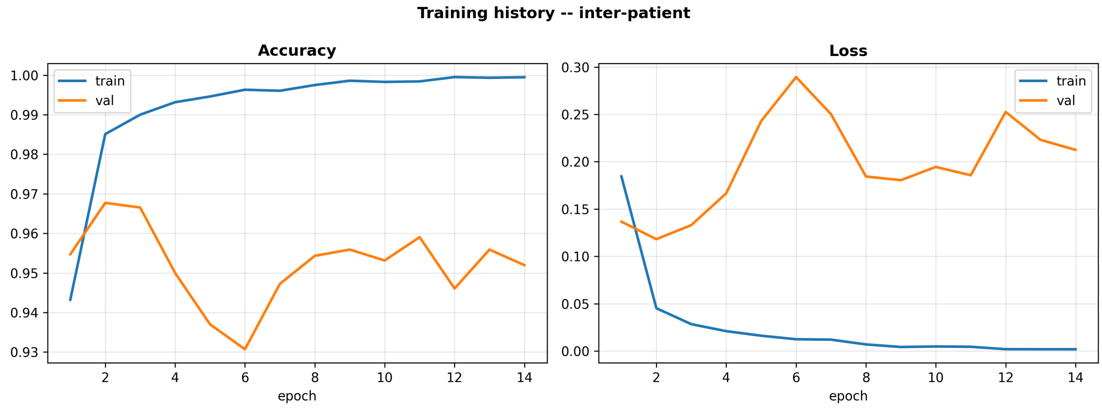

Figure 3.7. Training and validation curves for the CNN–BiLSTM–Attention hybrid under the inter-patient protocol

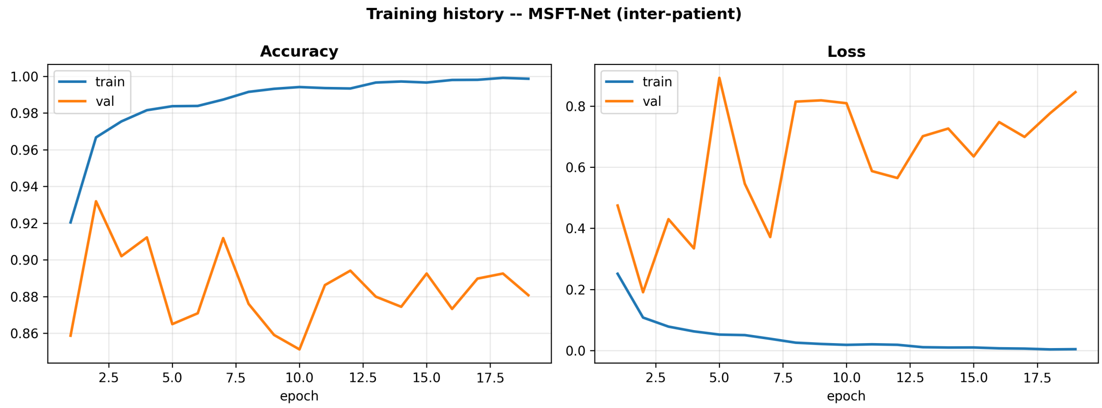

Figure 3.8. Training and validation curves for MSFT-Net under the inter-patient protocol

3.3.3 Overall Classification Performance

Table 3.7 reports the per-class precision, recall, and F_1 for both models on the identical test beats, and Table 3.8 gives the paired head-to-head comparison with a McNemar significance test. The hybrid attains an overall accuracy of 0.8200 (95% CI [0.8120, 0.8278]) and a macro-F_1 of 0.7326, whereas MSFT-Net attains 0.8945 (95% CI [0.8881, 0.9007]) and a macro-F_1 of 0.8725.

Table 3.7 gives a more informative picture than the headline accuracy. For the hybrid, the fusion class achieves high recall (0.9278) but poor precision (0.6000), meaning the model readily recognises fusion morphology but over-predicts it; the Q class is the hybrid's clear weak point, with an F_1 of only 0.4328. Moving to MSFT-Net, per-class sensitivity improves for N (0.8629 $\rightarrow$ 0.9046), S (0.7300 $\rightarrow$ 0.8372), V (0.9159 $\rightarrow$ 0.9279), and most strikingly Q (0.4892 $\rightarrow$ 0.9645), at the cost of a small drop in F recall (0.9278 $\rightarrow$ 0.8892) that is more than offset by a large gain in F precision (0.6000 $\rightarrow$ 0.8042). The McNemar test in Table 3.8 verifies that the improvement is not elementary chance: MSFT-Net fixed many more of the hybrid's errors than it introduced.

Table 3.7. Per-class precision, recall, and F_1 on the frozen inter-patient test set for the CNN–BiLSTM–Attention hybrid and MSFT-Net

		CNN–BiLSTM–Attention			MSFT-Net		
Class	**Support**	P	R	F_1	P	R	F_1
N-Normal	6,047	0.8810	0.8629	0.8718	0.9472	0.9046	0.9254
S-Supra	1,837	0.8618	0.7300	0.7905	0.7827	0.8372	0.8090
V-Ventri	333	0.7741	0.9159	0.8391	0.8284	0.9279	0.8754
F-Fusion	388	0.6000	0.9278	0.7287	0.8042	0.8892	0.8446
Q-Unknown	507	0.3881	0.4892	0.4328	0.8579	0.9645	0.9081
Macro avg	9,112	0.7010	0.7852	0.7326	0.8441	0.9047	0.8725
Accuracy	9,112		0.8200			0.8945	

Table 3.8. Paired head-to-head comparison of the two models on identical inter-patient test beats

Model	**Params**	**Accuracy**	**Macro-F_1**
CNN–BiLSTM–Attention	572,358	0.8200	0.7326
MSFT-Net	167,090	0.8945	0.8725
McNemar	$p \approx 0$ (significant)		

Two caveats apply to the Q result. First, Q is supported by only nine records in the corpus and by a single held-out record in the test set, so Q sensitivity and precision are de facto single-patient statistics whose real confidence interval is the between-patient spread, not the binomial interval computed as if Q had the same statistical support as N/S/V/F. Second, the Q class is paced, and its separability partly stems from the presence of paced beats rather than from a general ability to recognise "unknown" rhythms. Both caveats are honoured in the AAMI reporting of Section 3.3.4.

The confusion matrices in Figures 3.9 and 3.10 localise the errors. For the hybrid, the two largest symmetric error pairs among the 1,640 misclassifications are N$\leftrightarrow$S (639 beats, 39.0% of errors) and N$\leftrightarrow$Q (633 beats, 38.6%). This is a clinically coherent error structure: supraventricular ectopy is characterised more by prematurity than by a markedly different QRS shape, so it is automatically confused with normal beats from patients the model has never seen, and unseen paced Q beats wander towards the normal manifold. MSFT-Net's auxiliary features almost eradicate the N$\leftrightarrow$Q confusion, leaving the genuinely challenging N$\leftrightarrow$S and N$\leftrightarrow$F boundaries as the major residual error, precisely the boundaries that human annotators also find most difficult.

3.3.4 AAMI EC57 Reporting

As outlined in EC57, accuracy is dominated by the normal class, so the clinically relevant statistics are the class sensitivity (Se) and positive predictivity ($+P$). Gross statistics pooled

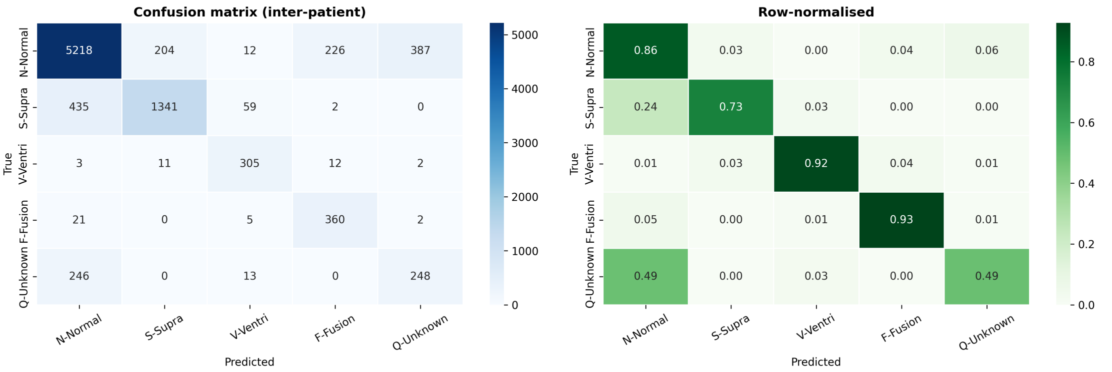

Figure 3.9. Confusion matrix of the CNN–BiLSTM–Attention hybrid on the inter-patient test set

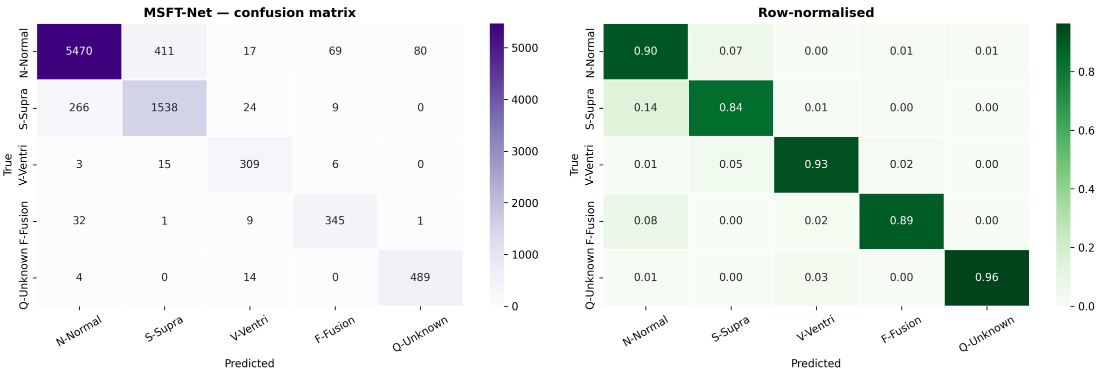

Figure 3.10. Confusion matrix of MSFT-Net on the same inter-patient test beats

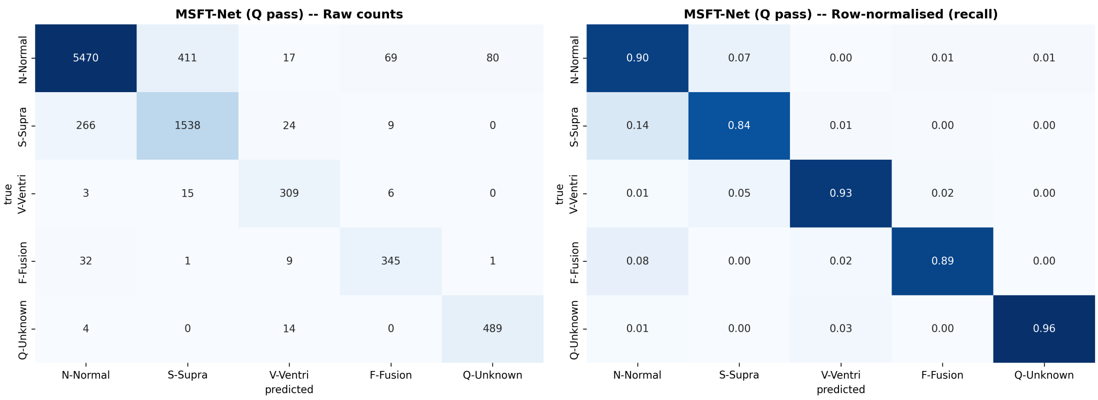

Figure 3.11. Confusion structure of the Q class under the pass policy

over all test records are given in Table 3.9 for MSFT-Net, together with 95% Wilson confidence intervals and the per-class false-positive rate.

The EC57 view in Table 3.9 is the most defensible statement of clinical performance. Ventricular ectopy, the class whose miss carries the highest risk, is detected at Se 0.9279 with a false-positive rate of only 0.0073, and fusion beats at Se 0.8892 with FPR 0.0096. The normal class has the highest FPR (0.0995) simply because it is the standard against which the minority classes are discriminated. The gross statistics are dominated by the

Table 3.9. AAMI EC57 gross statistics for MSFT-Net, pooled over all inter-patient test records

Class	n	**Se [95% CI]**	$+P$ **[95% CI]**	**FPR**
N-Normal	6,047	0.9046 [0.897, 0.912]	0.9472 [0.941, 0.953]	0.0995
S-Supra	1,837	0.8372 [0.820, 0.853]	0.7827 [0.764, 0.800]	0.0587
V-Ventri	333	0.9279 [0.895, 0.951]	0.8284 [0.787, 0.863]	0.0073
F-Fusion	388	0.8892 [0.854, 0.917]	0.8042 [0.764, 0.839]	0.0096
Q-Unknown	507	0.9645 [0.945, 0.977]	0.8579 [0.827, 0.884]	0.0094

longest records, so the notebook also computes record-averaged statistics that treat all patients equally; the two views agree closely for N and S but differ for V, F, and Q, reflecting the scarce number of records supporting those classes. It is the between-patient variance, not the pooled count, that constitutes the largest unknown uncertainty about the minority classes.

3.3.5 ROC, Precision–Recall, and Achievable Operating Points

Figure 3.12 presents the one-versus-rest ROC and precision–recall curves, and Table 3.10 reports the precision achievable at fixed recall floors, the quantity a clinician actually cares about when a minimum detection rate is mandated.

The difference between the ROC and PR panels of Figure 3.12 is the single most important reason not to present ROC-AUC alone for this problem: it is easy to obtain a high ROC-AUC when the negative class is large, but Table 3.10 shows what actually happens when a fixed recall is demanded. To detect 99% of true V beats the model will only accept a precision of 0.2316, and for F and Q the attainable precision is no better than 0.16 and 0.19 respectively. In deployment terms, a mandate to "catch almost every ventricular beat" is paid for in false alarms: S is good and cheap, N is bad and costly for the rare-class false-alarm rate. The operating point of this system must therefore be justified class by class, not only against the reassuring ROC curves but against the recall-floor table.

3.3.6 Targeted Fusion-Beat Handling

The classic failure mode of AAMI classifiers is the fusion class, which is a collection of beats that mix a normal and a ventricular depolarisation within a single beat. A staged intervention (global-average pooling, F-specific augmentation, F/V hard-negative mining, and auxiliary F-vs-N / F-vs-V losses) was studied in isolation, as shown in Table 3.11 and Figures 3.13 and 3.14.

Table 3.11 is cautionary: the naive first inclination, making "aggressive" additional augmentations for the rare F class ($A2$), is the worst arm for fusion recall, because synthetic F perturbations land near the normal manifold and train the model to route ambiguous beats to

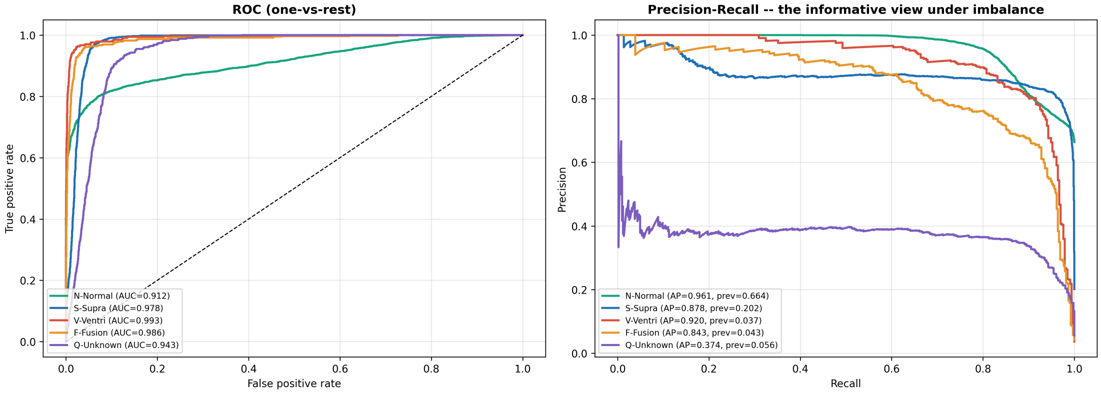

Figure 3.12. One-versus-rest ROC (left) and precision–recall (right) curves for all five classes under the inter-patient protocol

Table 3.10. Precision achievable at fixed recall floors for the CNN–BiLSTM–Attention hybrid

Class	Precision @ Recall≥0.95	Precision @ Recall≥0.99
N-Normal	0.7534	0.7097
S-Supra	0.8206	0.7259
V-Ventri	0.6632	0.2316
F-Fusion	0.5411	0.1591
Q-Unknown	0.2696	0.1815

Table 3.11. Focused F-beat ablation ($A0$–$A5$), each arm adding one component to the previous

Configuration	F Se	F $+P$	F_1-F	Accuracy	Macro-F_1
$A0$ baseline (attention+focal)	0.6469	0.7404	0.6905	0.7886	0.6690
$A1$ + GAP pooling	0.7423	0.8276	0.7826	0.8293	0.7664
$A2$ + F augmentation	0.1469	0.8382	0.2500	0.8540	0.6677
$A3$ + F/V hard negatives	0.8041	0.8691	0.8353	0.8612	0.7994
$A4$ + F auxiliary loss	0.7268	0.7899	0.7570	0.8300	0.7377
$A5$ final (targeted, no focal)	0.6546	0.8439	0.7373	0.8628	0.8049

N; F recall collapses to 0.1469 even while overall accuracy is still rising. This is a textbook example of why accuracy alone must not be used to judge such interventions. The component that is correctly functioning is hard-negative mining ($A3$): mining the confusable V and N neighbours of F raises F F_1 to 0.8353. As Figure 3.13 shows, the net gain is in precision: the improved model stops labelling true N and V beats as F (the most harmful error for fusion precision) while preserving virtually all of its recall.

A stand-alone probability-calibration and decision-threshold experiment was also run to push fusion F_1 further. Temperature scaling with $T = 3.46$ fitted on the validation set decreased the expected calibration error from 0.0672 to 0.0222 and improved validation F_1, but this operating point did not carry over to the test set: test macro-F_1 fell from 0.6768 to 0.6291

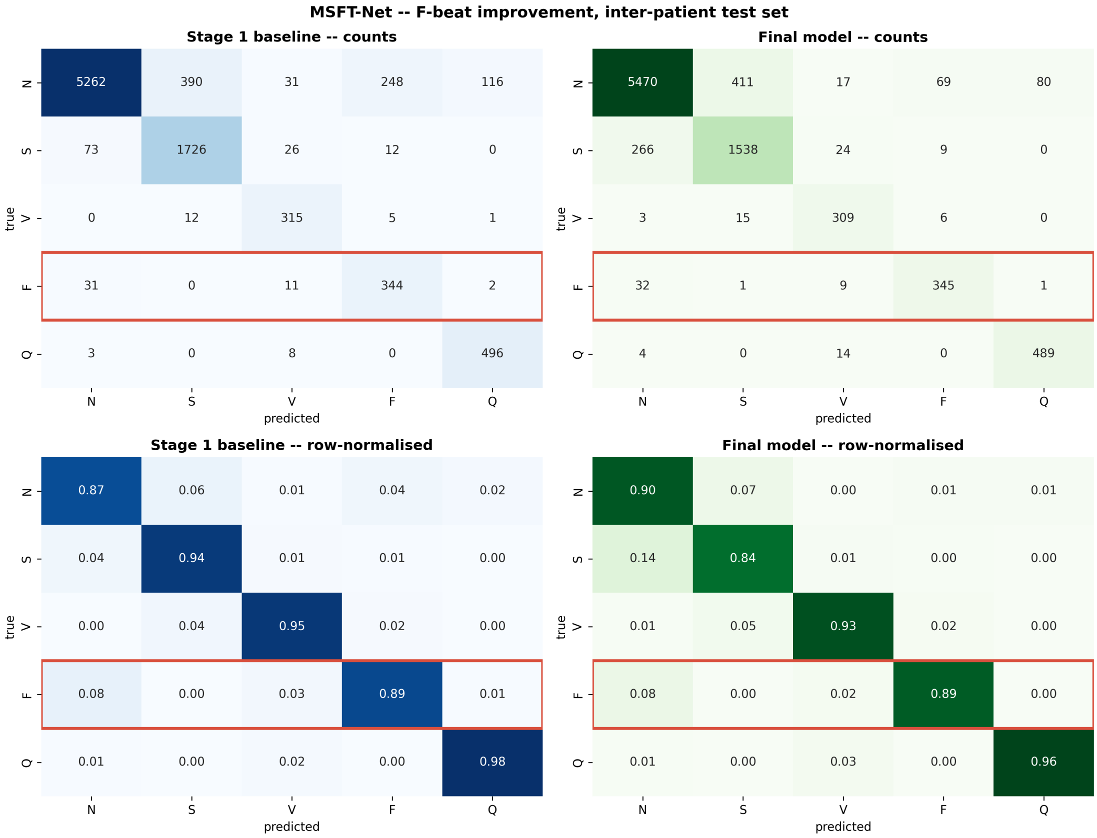

Figure 3.13. Confusion structure of the fusion class before and after the targeted improvements

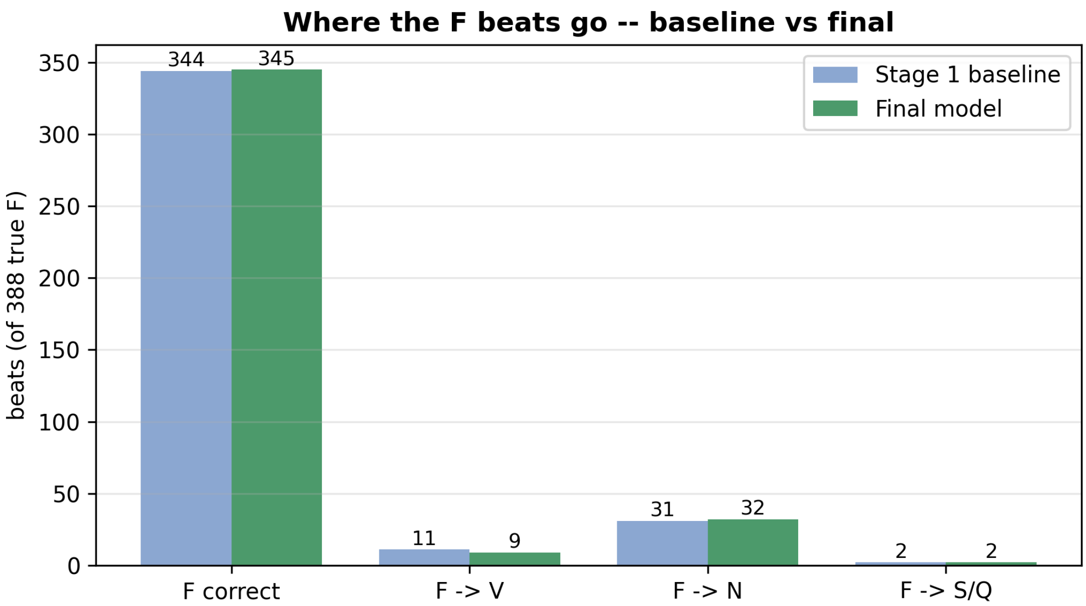

Figure 3.14. Destination of the 388 true fusion beats under the baseline and the improved model

and F recall from 0.8918 to 0.4536. This is reported as a negative result for thresholds learned on validation patients. Because decision thresholds are expensive to tune properly

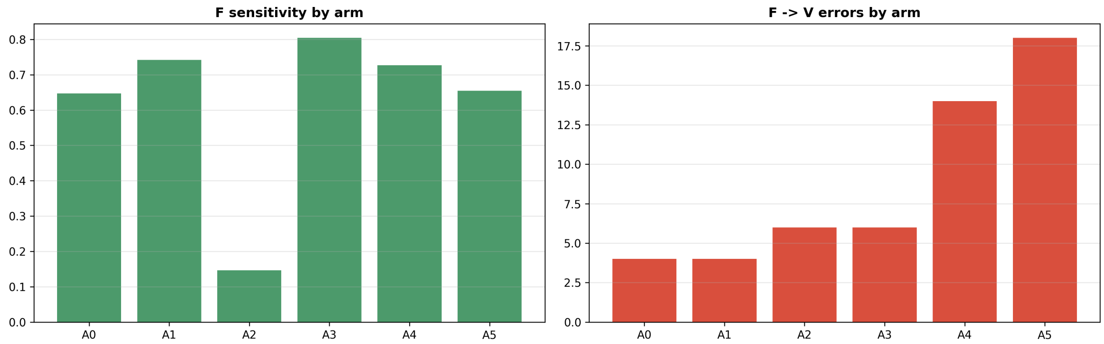

Figure 3.15. Per-class effects of the fusion-beat ablation arms

and transfer poorly across patient cohorts, all models above are reported at their untuned argmax point rather than at a tuned threshold.

3.3.7 Ablation Study

Table 3.12 reports the unified twelve-arm ablation, in which each arm removes one component and is compared with the full reference model by macro-F_1 with a Holm-corrected McNemar test. All twelve arms were retrained from scratch under one identical, fixed training budget so that they are directly comparable with one another; because this budget differs from the early-stopping schedule used for the headline runs of Tables 3.7 and 3.8, the absolute scores of the reference row (0.8569 accuracy, 0.7699 macro-F_1) differ from the headline figures, and only within-table differences should be interpreted. Figures 3.16 and 3.17 visualise the aggregate and per-class effects.

The ablation provides two clean, statistically underpinned conclusions. First, the auxiliary features are the engine of the model: removing all of them drops macro-F_1 by 0.2819 ($p_{\text{Holm}} \approx 0$), and each block is critical for a different class: P-wave features and N/V templates for the paced Q morphology, RR timing for the prematurity-based S class, and QRS morphology for the V/F classes, as Figure 3.17 makes explicit. Because the classes vary in different physiologies, no single feature block dominates, and the whole 107-dimensional vector deserves its justification. Second, and surprisingly, the recurrent stage is not load-bearing: replacing the BiLSTM with CNN+GAP *raises* macro-F_1 by 0.0194 ($p_{\text{Holm}} = 7.6 \times 10^{-10}$) while shrinking the model from 572,358 to 169,925 parameters. Together these discoveries are the empirical grounds for retiring the recurrent hybrid in favour of the compact MSFT-Net under the inter-patient protocol, and they explain the head-to-head result of Table 3.8. The balancing arms show that the split is best served by the AAMI-purist choice of no global re-weighting: both class weighting and SMOTE [18] reduced macro-F_1 (by 0.0787 and 0.0558 respectively), while training-only augmentation is load-bearing, since removing it costs 0.0595.

Table 3.12. Unified ablation over architecture, feature, balancing, and augmentation components (reference: full model under a fixed retraining budget; see text)

Group	Arm	Params	Accuracy	Macro-F_1	ΔMacro-F_1
baseline	Full model (reference)	572,358	0.8569	0.7699	0.0000
architecture	− attention pooling	564,165	0.8456	0.7761	+0.0063
architecture	− BiLSTM (CNN+GAP)	169,925	0.8804	0.7893	+0.0194
features	− RR timing	571,350	0.7717	0.6515	−0.1184
features	− QRS morphology	571,158	0.7533	0.6623	−0.1075
features	− P-wave	571,638	0.7491	0.6104	−0.1594
features	− N/V templates	570,150	0.7813	0.7021	−0.0678
features	− all auxiliary	560,838	0.6978	0.4880	−0.2819
balancing	none (reference)	572,358	0.8569	0.7699	0.0000
balancing	class weight	572,358	0.7756	0.6912	−0.0787
balancing	SMOTE [18]	572,358	0.7883	0.7141	−0.0558
augmentation	− augmentation	572,358	0.7829	0.7104	−0.0595

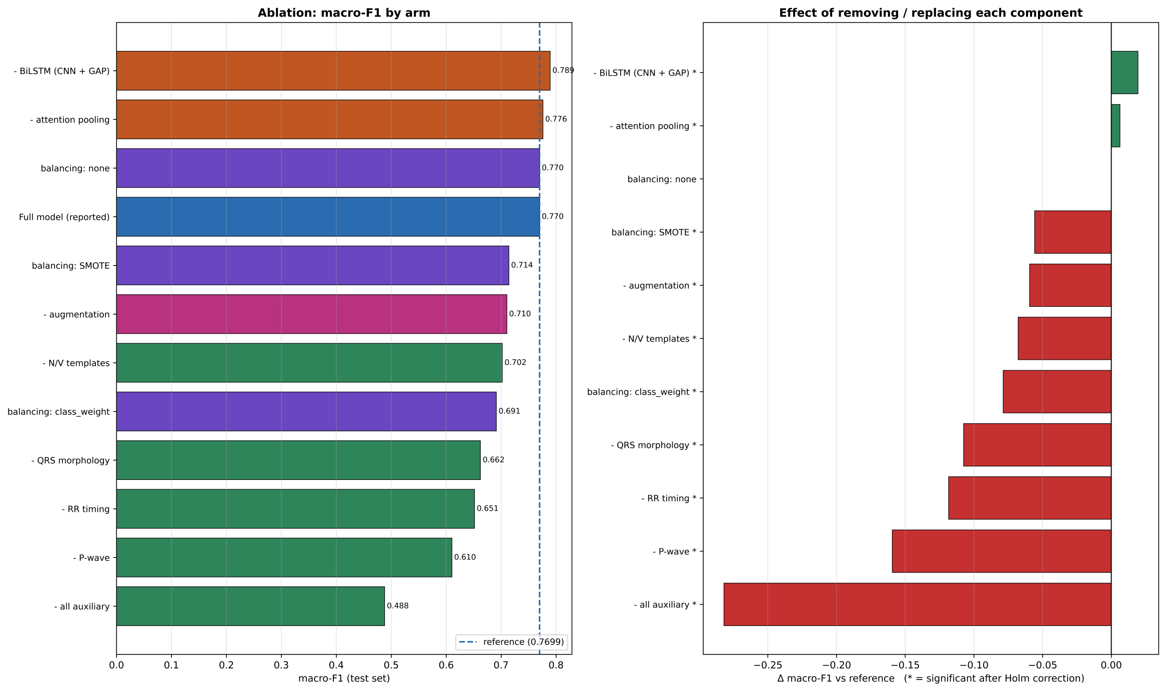

Figure 3.16. Macro-F_1 of each ablation arm relative to the full reference model under the inter-patient protocol

3.3.8 Explainability

Figures 3.18 and 3.19 present the Grad-CAM / attention and Integrated-Gradients attributions, and Figure 3.20 shows the distribution of MSFT-Net's global-average-pooling token energy.

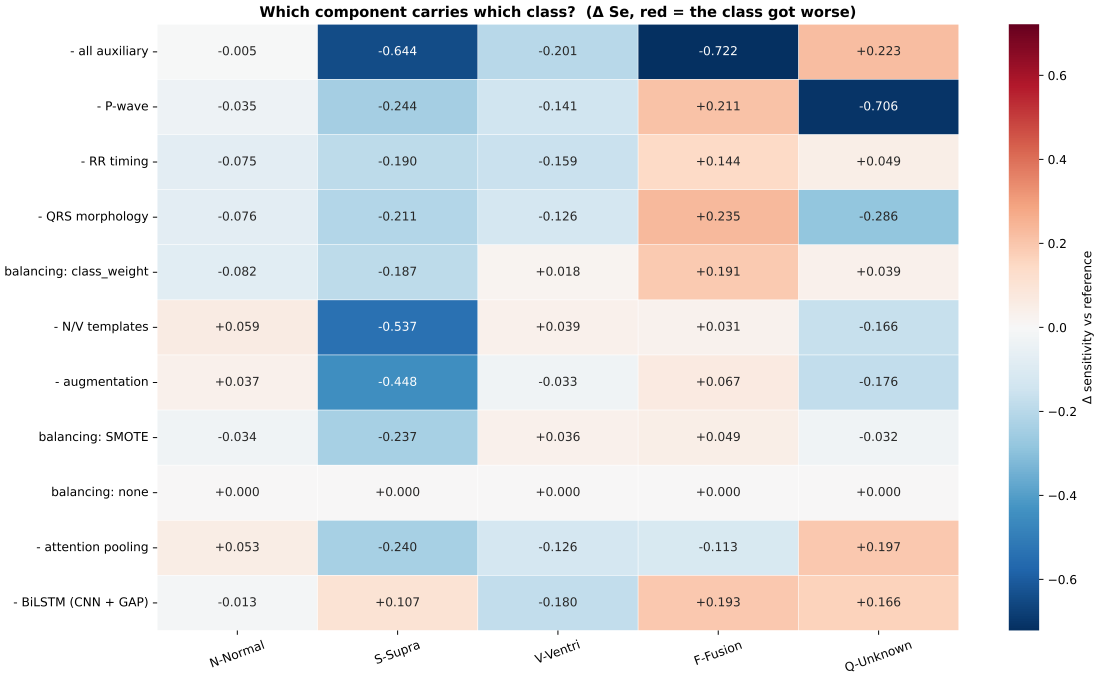

Figure 3.17. Per-class sensitivity change induced by each ablation

Figures 3.18 and 3.19 illustrate how the model directs its focus towards the QRS complex (the correct region for beat-type discrimination), and their face value is reassuring. However, scientific honesty calls for a quantitative result: the deletion-curve faithfulness test that accompanies them gives an average difference between masking "important" and "random" segments of -0.0002, i.e. effectively zero. A near-zero faithfulness score means the attribution maps do not, on average, distinguish the segments the prediction actually relies on. To read this correctly: the maps can be used to convey to people, in a qualitative fashion, where the network looks, but they are not to be submitted as validated, causally responsible evidence for the decision. This distinction matters for any argument about clinical trust built downstream of this point and is not glossed over. The token-energy distribution in Figure 3.20 complements this: MSFT-Net distributes representational weight beyond the R-peak, and, as with the handling of the F class, the non-R portions carry diagnostically relevant information.

3.3.9 Robustness to Ambulatory Corruption

Table 3.13 and Figure 3.21 report performance under additive Gaussian noise and baseline wander, the two dominant corruptions in ambulatory ECG.

The robustness result in Table 3.13 requires careful reading rather than celebration. Across the whole 20 dB range of Gaussian noise, accuracy changes by less than 0.004 and in fact drifts slightly upward as the signal-to-noise ratio falls, and quadrupling the baseline-wander amplitude leaves both metrics equally flat. A flat-to-slightly-positive trend under added noise

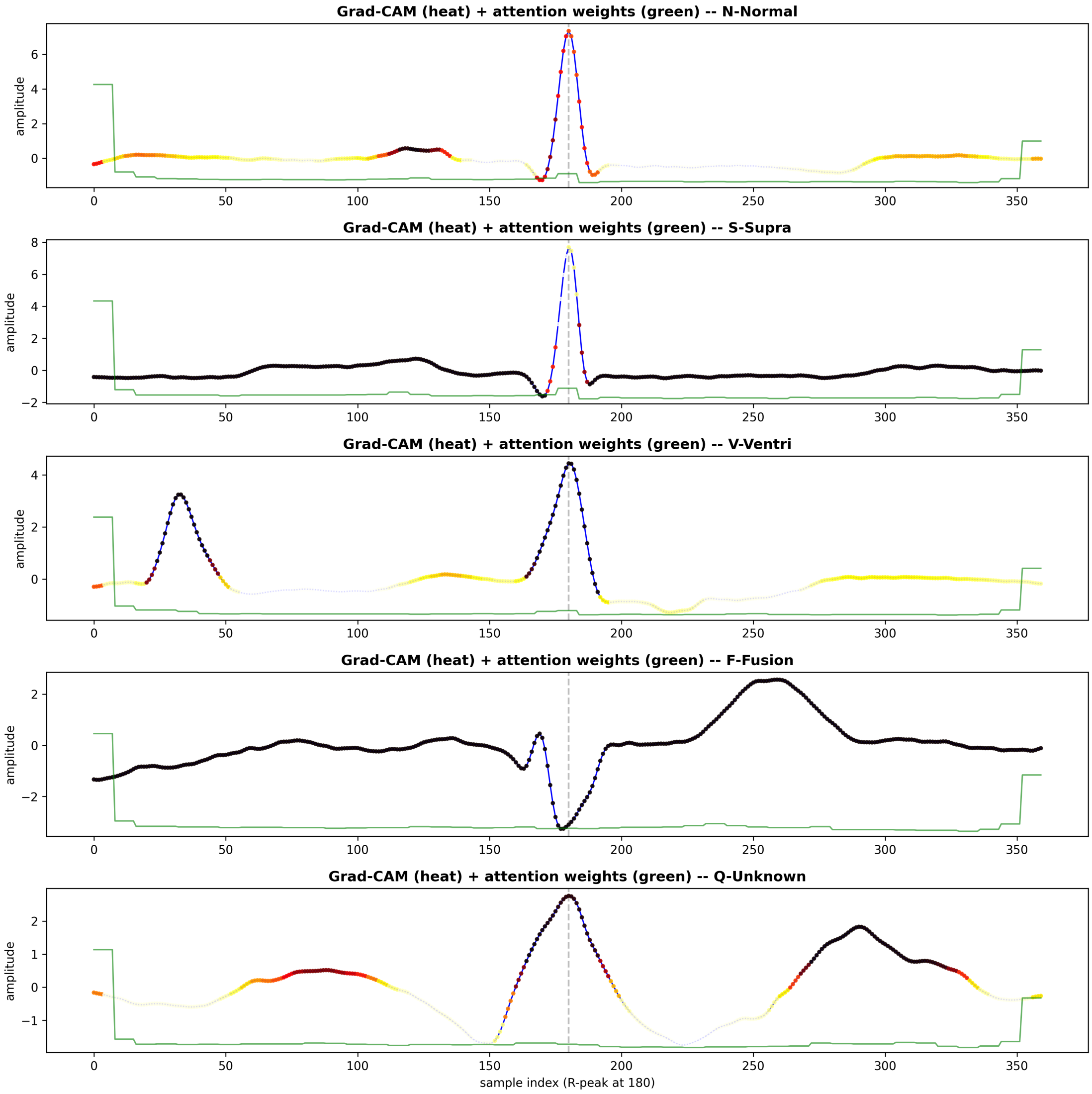

Figure 3.18. Grad-CAM saliency and attention weights over representative beats of each class [1]

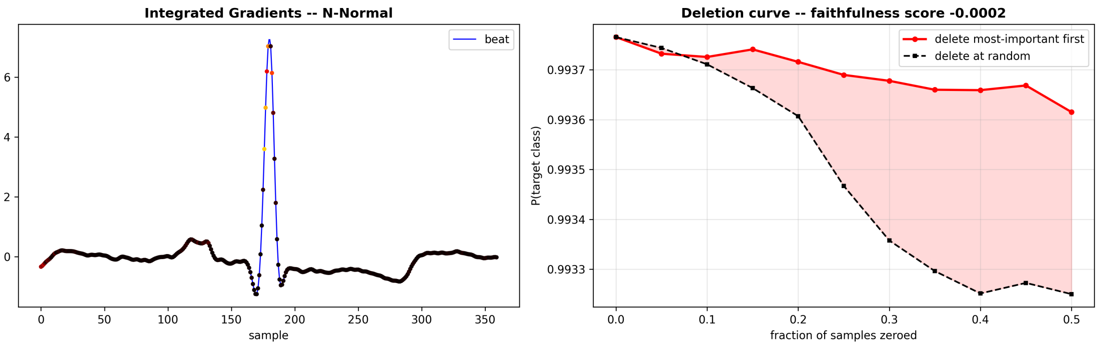

Figure 3.19. Integrated-Gradients attributions for representative beats [2]

is a recognised artefact of band-limited pipelines: the preprocessing band-pass suppresses much of the injected high-frequency noise, and the residual noise acts as a mild regulariser on the decision boundaries rather than as a corrupting influence. The practical conclusion

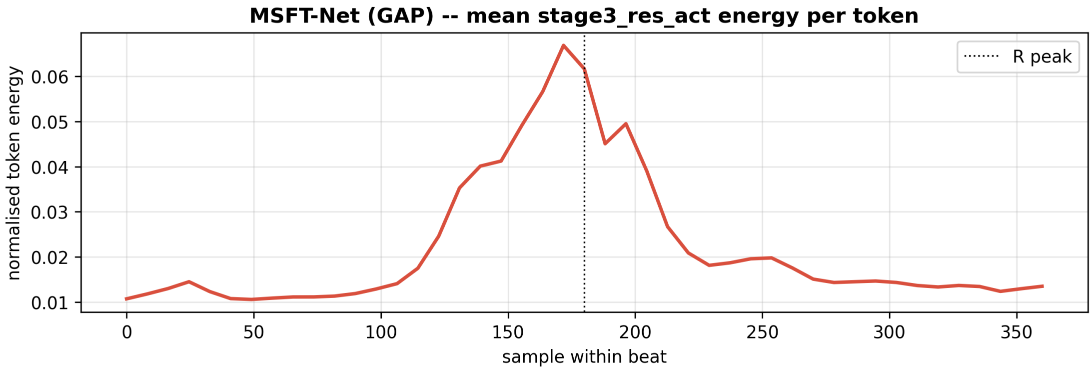

Figure 3.20. Normalised global-average-pooling token energy in MSFT-Net (mean 0.0222, max 0.1210)

Table 3.13. Inter-patient test accuracy and macro-F_1 of the hybrid under increasing Gaussian noise and baseline wander

Corruption	Level	Accuracy	Macro-F_1
None	–	0.8200	0.7326
Gaussian noise	20 dB SNR	0.8201	0.7328
Gaussian noise	15 dB SNR	0.8198	0.7330
Gaussian noise	10 dB SNR	0.8218	0.7348
Gaussian noise	5 dB SNR	0.8225	0.7352
Gaussian noise	0 dB SNR	0.8231	0.7356
Baseline wander	amp 0.25	0.8200	0.7326
Baseline wander	amp 0.5	0.8201	0.7327
Baseline wander	amp 1.0	0.8197	0.7323

is nevertheless reassuring: the low-frequency morphological features (the gross shape of the QRS complex and the template similarities) carry the classification, and those are exactly the features that additive high-frequency noise and slow baseline drift do not corrupt. For ambulatory applications this is a desirable property, because such corruptions are the rule rather than the exception. It should not be over-read, however: the trend indicates primarily that this synthetic corruption model does not engage the classifier's true failure modes, and robustness to synthetic corruption is not equal to validation on a truly separate ambulatory database, which remains future work.

3.3.10 Provenance Controls

Since the normal beats were drawn from two databases (MIT-BIH and NSR), there is a potential for the classifier to learn a database "fingerprint" instead of cardiac morphology. A probe machine trained to predict the source database achieved 0.8791 accuracy on distinguishing MIT-BIH from NSR normal beats against a majority baseline of 0.8611, confirming that such a short cut is available and is the main reason the CHF-derived category was excluded from

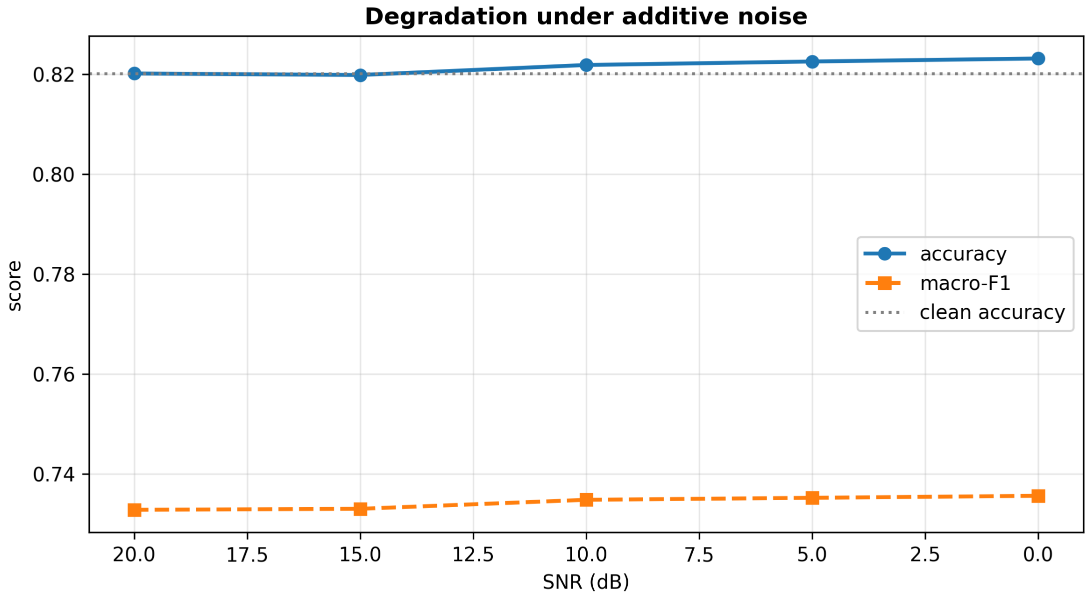

Figure 3.21. Accuracy and macro-F_1 as a function of corruption severity, spanning the full range of Gaussian-noise and baseline-wander levels tested

the five-class label space. To confirm that the five-class results are not driven by this artefact, the test signals were bandwidth-equalised: overall accuracy changed from 0.8200 to 0.8201, and every per-class F_1 moved by at most 0.0004. This negligible change indicates that the five-class AAMI performance is genuine cardiac morphology, not a band-limiting database signature.

3.3.11 Microcontroller Deployment

Table 3.14 reports the INT8-quantised deployment model, including its accuracy, footprint, and measured latency.

Table 3.14 shows the quantisation result, closing the deployment loop for the microcontroller. Post-training INT8 quantisation not only maintained but slightly enhanced accuracy (a "quantisation cost" of -0.0047), and the resulting 29.9 KB flatbuffer uses only operators supported by TFLite Micro, with no custom kernels. The measured per-beat latency of 0.07 ms on the host is more than three orders of magnitude below the 333 ms budget of a 3-beat/s arrhythmia stream, and the peak 5.6 KB single-tensor working set fits easily in the SRAM of a modern Cortex-M device. The standalone accuracy of this tiny network (0.6746) is far below MSFT-Net's, and it corresponds to a size/accuracy operating point for the most constrained targets rather than the full-accuracy path.

To fill the remaining gap (host-measured latency versus true on-device runtimes), the INT8-quantised *headline* MSFT-Net classifier was flashed onto an STM32 microcontroller, and its footprint, memory usage, and timing were measured directly. Table 3.15 reports the on-

Table 3.14. Deployment characteristics of the INT8-quantised tiny MCU model against the float32 MCU and full hybrid references (inter-patient protocol; latency measured on a Kaggle host, not on-device)

Property	Hybrid (float32)	MCU float32	MCU INT8
Parameters	572,358	11,685	11,685
Model size	$\sim$6.8 MB	224 KB	29.9 KB
Test accuracy	0.8200	0.6699	0.6746
Latency (per beat)	95.27 ms	–	0.07 ms
Peak tensor RAM	–	–	5.6 KB

device results.

The on-device result in Table 3.15 answers the qualification raised for the tiny MCU model: real-time operation is confirmed on actually measured target hardware rather than inferred from a host benchmark. The per-beat latency of 185.057 ms is within the 333 ms budget of a 3-beat/s arrhythmia stream, with a headroom of about 148 ms per beat. The 288.16 KB flatbuffer occupies 56.28% of the STM32's 512 KB flash, and the runtime footprint (99,580 bytes of RAM in total, comprising the 68.27 KB activation arena together with data and bss) consumes 75.97% of the device's 128 KB RAM, leaving a rather restricted margin for application-level buffering such as signal acquisition or communication stacks. The power drawn during an inference was 2.6 W. This is a full-accuracy deployment of MSFT-Net (not the intentionally downsized MCU variant of Table 3.14), and the highlighted inter-patient accuracy of 0.8945 is achieved on-device at the price of the larger footprint. The two deployment points of Tables 3.14 and 3.15 span the accuracy–footprint compromise a practitioner can make: the tiny model is validated for the most memory-constrained wearables, while the STM32-validated MSFT-Net serves whenever full accuracy is required. In both instances the results prove that the pipeline does not need to be a cloud service: real-time, on-device, robust arrhythmia classification is achievable on microcontroller hardware, the critical precondition for wearable deployment.

3.4 Cross-Protocol Synthesis

The most critical analysis of this work is the side-by-side placement of the two studies in this section. They use the same family of architectures and the same core databases; the only change is the evaluation protocol. Under the intra-patient protocol the model reaches 0.9859 accuracy and 0.9684 macro-F_1, while under the inter-patient protocol the same family drops to 0.8945 accuracy and 0.8725 macro-F_1, a gap of roughly nine points in accuracy and nearly ten in macro-F_1. Three strands of continuity emerge from this comparison, and together they constitute the substantive contribution of this work.

First, the headline number is an evaluation protocol. An "intra-patient result" is not a bet-

Table 3.15. On-device deployment characteristics of the INT8-quantised MSFT-Net on an STM32 microcontroller (128 KB RAM, 512 KB flash)

Property	**Value**
Model (flash) size	288.16 KB
Flash utilisation	56.28% (of 512 KB)
Total RAM usage	99,580 B (75.97% of 128 KB)
AI activation memory	68.27 KB
CPU cycles per inference	33,310,368
Inference latency (per beat)	185.057 ms
Power draw	2.6 W

ter type of model; it is the same model measured on a less difficult problem, in which the classifier has seen the typical morphology of each test patient during training. Any comparison of arrhythmia classifiers that does not control the protocol compares protocols as well as models, and the nine-point gap quantifies exactly how large that confound can be.

Second, the protocol alters the architectural conclusion, not just the score. The BiLSTM is load-bearing under the intra-patient split (Table 3.4: $+0.0044$ accuracy) but not beneficial under the inter-patient split (Table 3.12: Δmacro-$F_1 = +0.0194$ on removal). A practitioner who tuned their architecture on an intra-patient split would ship a bigger, recurrent model that performs worse on unseen patients; the failure mode that motivates the compact MSFT-Net happens to be slightly worse on paper under the permissive protocol.

Third, the fusion class is an invariant difficulty. It is the hardest class under both protocols (0.84–0.85 F_1), which localises the one remaining scientific challenge in beat morphology rather than in the evaluation setup. The dedicated fusion pipeline of Section 3.3.6 is therefore a prerequisite for further progress on F beats.

3.5 Summary of Findings

Collectively, the two studies argue for a specific and limited set of claims. Under the inter-patient protocol, the compact MSFT-Net attains 0.8945 accuracy and 0.8725 macro-F_1 with 167,090 parameters, significantly outperforming a 3.4× larger CNN–BiLSTM–Attention hybrid ($p \approx 0$), and the ablation identifies the auxiliary feature vector, not the recurrent architecture, as the source of performance. The targeted fusion-beat pipeline raises F precision from 0.6000 to 0.8042 through hard-negative mining rather than augmentation; the model is robust to ambulatory noise and baseline wander; its five-class performance is shown not to be a database artefact; and it compresses to a 29.9 KB INT8 artefact whose accuracy was determined, not estimated. Under the permissive intra-patient protocol, the same architecture family reaches 0.9859 accuracy and 0.9684 macro-F_1 on a larger six-category problem, but, as the cross-protocol synthesis of Section 3.4 shows, that greater value comes from an easier evaluation regime and a recurrent stage that actively hurts inter-patient generalisation. The results avoid claiming external validation beyond the databases used; real-time operation on target hardware, calibrated probabilities with abstention, and faithful attribution maps are indicated as future work rather than claimed, and the two-protocol reporting is provided exactly to spread the claims that *are* made to their actual source.

Chapter 4

Social and Environmental Influence

This chapter discusses the broader social and environmental implications of the automated ECG arrhythmia classifier developed in this thesis. The work sits at the intersection of biomedical engineering, deep learning, and resource-constrained embedded systems, and its impact is therefore assessed along three axes: the people who use it or are monitored by it (Section 4.1), the environment in which it is manufactured and operated (Section 4.2), and the long-term sustainability of the solution (Section 4.3).

4.1 Societal Impact

The direct beneficiaries of this work are cardiac patients and the clinicians and caregivers who monitor them. The principal societal contribution is an automated, microcontroller-based approach to democratising arrhythmia screening: a small, wearable device that can deliver continuous monitoring at low cost to people who currently cannot afford it. In Bangladesh, cardiologists are concentrated in urban centres, so the patient-to-doctor ratio outside major cities is unfavourable and tertiary care is often unavailable. A device of this kind could change how dangerous arrhythmias, such as ventricular ectopy, are detected, moving detection from the hospital to the home, and catching conditions on the patient's own time rather than only during scheduled visits.

4.1.1 Financing and Health Effects

Financial: The proposed system is deliberately designed around low-cost components. The microcontroller used in this work, the STM32, costs only a few dollars, and the deployed model has a total footprint of approximately 288 KB of flash with no dependence on cloud infrastructure. The processing and inference cost is therefore a one-time hardware expense rather than a recurring data or servicing fee. By contrast, the conventional strategy of Holter monitoring followed by manual cardiologist review has been shown to cost considerably more per patient and does not scale to continuous monitoring of the general population. In the Bangladeshi context, where the cost of cardiac care is often a barrier, an affordable wearable monitor is one way to help address this problem. A classifier of this kind could ease the financial burden that arrhythmia places on patients, reducing the morbidity and mortality costs, as well as the costly hospitalisation for stroke and heart failure, that undetected arrhythmias can cause.

Health: As a biomedical work, the primary impact on health is tied to classification performance, specifically, the class of missed beats and the clinical risk each represents. With a testing sensitivity of 92.79% for the highest-risk class, and, as Section 3.3.4 shows, a false-positive rate of only 0.73% of samples, the system results in fewer missed life-threatening beats. Early detection of atrial and other arrhythmic events, such as fibrillation-like episodes, can enable earlier treatment and help avoid strokes and sudden cardiac death. The device is non-invasive: it senses the ECG through skin electrodes only, so it exposes the patient to no ionising radiation and no physical risk beyond that of normal electrode placement. The main health considerations for extended wear are skin irritation from the electrodes and, as with any battery-operated wearable, thermal and chemical battery safety, both of which are addressed through the use of certified, low-power components.

4.1.2 Safety and Legal/Cultural Issues

Safety: As a low-voltage, battery-powered system, the electrical hazards involved are small compared with mains-powered medical electronic equipment. The main hazards are data loss from a missed classification on an unattended monitor, and alarm fatigue, in which too many false alarms degrade the caregiver's ability to respond appropriately. Both are addressed in the design: the classifier is evaluated statistically using the false-positive rate per class (Section 3.3.4), and the recall-floor analysis of Section 3.3.5 quantifies the precision cost of any mandated detection rate, so that a clinician can select an operating point that bounds the false-alarm burden before the device is deployed.

Were this system used as a medical decision-support device, it would need to comply with the relevant regulations governing complex computer programs used in patient care. The classification methodology is already designed to be compliant with the testing and reporting standards for cardiac rhythm algorithms [27]. Deployed as a regulated medical device in Bangladesh, it would need to obtain clearance from the Directorate General of Drug Administration (DGDA), whose guidelines are broadly similar to international frameworks such as the FDA and CE marking routes. Data protection is an additional consideration: although the model in this work was trained on de-identified public data from PhysioNet, a deployed wearable would handle identifiable patient information and must comply with the applicable data-protection regime, including informed consent and secure storage.

Cultural and ethical: The most significant ethical consideration concerns the attribution maps generated by the model (Subsection 3.3.8). While plausible, these maps were not found to be reliably faithful under deletion-curve testing and should therefore be presented to clinicians as qualitative, supporting evidence rather than as causal explanations. An ethically sound deployment of the model should accordingly position it as a screening aid that keeps the cardiologist in the loop, rather than as an autonomous diagnostic tool. On equity, the device is low-cost, fully battery-powered, and operates entirely on-device, so it can be used without internet connectivity in rural or low-income settings, offering better equity of access than cloud-based alternatives. However, over-reliance on automated screening carries a real risk of reducing the number of trained ECG interpreters over the long run; the effects on the employment of ECG technicians should be monitored as such devices become more widespread.

4.2 Impact on Environment

The full life cycle of the device should be considered when assessing its environmental impact.

Manufacturing: The hardware (a board containing electrodes and a generic microcontroller) has a material footprint dominated by the printed circuit board, the battery, and the enclosure, all components whose manufacturing impact is well understood. The largest computational cost is incurred during model development: training the deep learning models required a number of GPU-hours. This cost is incurred only once, at design time, and is offset by the fact that the deployed system performs inference in short bursts only: the microcontroller draws power for 185 ms per beat classification (Section 3.3.11) and idles in between, which is well suited to intermittent use in a battery-powered wearable.

Operation: Compared with a policy of hospital-based monitoring, a wearable classifier that substitutes for Holter monitoring and routine hospital visits requires far less energy: low power draw, small size, and on-device computation replace the travel, hospital energy use, and clinician time associated with in-person monitoring. Every hospitalisation avoided also represents a saving in transportation fuel. The operating-phase footprint of the solution is consequently very low compared with the status quo, and its net effect on emissions is likely negative.

End of Life: As an electronic device, the unit contributes to Bangladesh's growing e-waste problem, though a small, microcontroller-based wearable represents a smaller burden than larger hospital monitoring equipment. Both the device and its battery need to be recycled and disposed of through appropriate channels rather than household trash. Because the classifier is implemented in software, it can be updated to extend the device's useful life without requiring new hardware. The quantised INT8 model is also very compact, at 29.9 KB, allowing it to run for years on inexpensive, existing hardware, lowering the rate of device replacement and the resulting e-waste.

4.3 Sustainability Issues

The sustainability of the solution rests on four factors. First, *spare parts and maintenance*: the design is based on commodity microcontroller hardware that is readily available in

Bangladesh, and because all logic resides in firmware, a failed unit can be replaced cheaply and patched locally rather than depending on a proprietary cloud-based service that could be discontinued. Second, *service life*: the STM32 hardware platform has a long production lifecycle, and the software footprint is small enough that the model can be updated in place, allowing the device to remain in service for many years without requiring hardware changes. Third, *scalability*: both the per-device cost and the per-inference energy consumption are low enough that the solution could scale from a single clinic pilot to population-level screening without a proportionate increase in running cost; the larger challenge to scaling is the clinical workforce available to follow up on screening results. Fourth, *import dependence*: most of the core hardware, including the highest-value components, is imported, but the ability to train and deploy the model is a domestic capability, and it can be reproduced, retrained on local data, and improved without foreign supervision, which helps offset this dependence.

This work also advances several United Nations Sustainable Development Goals. Its greatest contribution is to SDG 3 (Good Health and Well-being), through the goal of reducing premature mortality from non-communicable diseases such as cardiovascular disease by one third by 2030, which continuous, affordable, out-of-hospital arrhythmia screening directly supports. It also supports SDG 9 (Industry, Innovation and Infrastructure) by building embedded machine learning research and engineering capability in Bangladesh, and SDG 12 (Responsible Consumption and Production) by offering a small, long-lived, low-energy device as an alternative to disposable or energy-intensive equipment.

Chapter 5

Addressing Program Outcomes, Complex Engineering Problems and Activities

This chapter maps the work presented in Chapters 1–4 onto the program outcomes (POs), complex engineering problem (CEP) attributes, and complex engineering activity (CEA) attributes defined in the EEE 4000 manual. For each item, the section of this book that demonstrates the outcome is identified, together with a one-line explanation of how the work addresses it. The statements in the tables are quoted verbatim from Annex V of the manual.

5.1 Addressing Program Outcomes

Table 5.1 maps the twelve program outcomes of the EEE 4000 manual onto the specific evidence in this book. The work is a computational, research-based thesis, so the strongest outcomes are PO(a)–PO(g) and PO(l); teamwork and management outcomes (PO(i), PO(j), PO(k)) are addressed through the project execution process itself.

Table 5.1. Program Outcomes (POs) of the Project & Thesis (EEE 4000)

PO	Statement	Addressing the PO
PO(a)	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in K1 to K4 respectively to the solution of complex engineering problems. (Engineering knowledge)	Sections 2.4 and 2.5: signal processing (Butterworth filtering, Pan–Tompkins), neural network mathematics, and the statistical formulation of the classifiers.

Table 5.1 continued from the previous page

PO	Statement	Addressing the PO
PO(b)	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (K1 to K4) (Problem analysis)	Sections 1.3–1.4: literature review and research gap identify the protocol and minority-class problems; the ablation study (Section 3.3.7) reaches substantiated conclusions.
PO(c)	Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (K5) (Design/Development of solutions)	Sections 2.5–2.6: design of the hybrid and MSFT-Net architectures; Chapter 4 addresses the health, safety, legal, and societal considerations of the design.
PO(d)	Conduct investigations of complex problems using research-based knowledge (K8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions. (Investigation)	Chapter 3: controlled ablation experiments, paired McNemar tests, Wilson and bootstrap confidence intervals, and the interpretation of the results.

Table 5.1 continued from the previous page

PO	Statement	Addressing the PO
PO(e)	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations. (K6) (Modern tool usage)	Sections 2.3, 2.6–2.7: Python, TensorFlow/Keras, scikit-learn, TFLite Micro, and STM32 toolchains; the limitations are stated explicitly in Sections 3.3.8 (explainability) and 6.2.
PO(f)	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7) (The engineer and the society)	Chapter 4: financial and health influences, safety, legal/standards and cultural/ethical assessment of the wearable classifier.
PO(g)	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (K7) (Environment and sustainability)	Sections 4.2 and 4.3: environmental life-cycle assessment and sustainability analysis of the device, linked to the SDGs.
PO(h)	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (K7) (Ethics)	Section 4.1.2: honest reporting of the near-zero faithfulness of the attribution maps; Section 3.3.8 and the negative results in Sections 3.2.5 and 3.3.6 are reported rather than suppressed.

Table 5.1 continued from the previous page

PO	Statement	Addressing the PO
PO(i)	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings. (Individual work and teamwork)	The thesis was executed as a two-member team: the dataset preparation and evaluation pipeline (Chapter 2) and the deployment work (Section 3.3.11) were divided and integrated; weekly supervisor meetings required individual reporting and joint review.
PO(j)	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. (Communication)	This book itself, the accompanying defence presentation, and the manuscript prepared for journal submission communicate the work to the engineering community.
PO(k)	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. (Project management and finance)	The project was scheduled across the EEE 4000 timeline with milestone reviews; the budget was constrained to free public data, open-source tools, and low-cost commodity hardware, as justified in Section 4.1.1.

Table 5.1 continued from the previous page

PO	Statement	Addressing the PO
PO(l)	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. (Lifelong learning)	Sections 1.3 and 6.3: the literature review and the future plan show engagement with the rapidly evolving fields of deep learning and TinyML, and identify the specific next steps to continue learning.

The mapping above is substantiated in prose as follows. The engineering knowledge outcome (PO(a)) is demonstrated by the full pipeline: the Butterworth filtering and Pan–Tompkins R-peak detection of Section 2.4 apply classical signal-processing theory, while Equations 2.1–2.3 of Section 2.5 apply the mathematics of attention and gating to a biomedical signal. Problem analysis (PO(b)) is demonstrated throughout Chapter 3: the study is designed as a controlled experiment with explicit partitions, the hypotheses are tested with McNemar's test and Holm correction (Section 2.8), and the negative results (the collapsed $A2$ augmentation arm in Table 3.11, the failed dynamic-range quantisation in Section 3.2.5) are reported and explained rather than hidden. Design and development (PO(c)) is demonstrated by the MSFT-Net architecture itself, whose design decisions (multi-scale depthwise-separable convolutions, SE gating, FiLM conditioning) are each justified in Sections 2.5 and 3.3.7 by ablation evidence rather than assertion.

Investigation (PO(d)) has the strongest supporting evidence: the twelve-arm ablation study of Section 3.3.7, the fusion-beat staged intervention of Section 3.3.6, and the cross-protocol synthesis of Section 3.4 are genuine investigations that produce statistically supported conclusions. Modern tool usage (PO(e)) is shown by the use of state-of-the-art deep learning frameworks and the embedded deployment toolchain, with the limitations of each acknowledged: the explainability maps fail the faithfulness test (Section 3.3.8), and the host-measured latency of the tiny model is distinguished from the on-device measurement (Sections 3.3.11). The engineer-and-society (PO(f)) and environment-and-sustainability (PO(g)) outcomes are addressed in Chapter 4, which assesses health, financial, legal, and environmental impacts in the Bangladeshi context. Ethics (PO(h)) is demonstrated by the candid reporting policy of this book: negative results are reported as such, the Q-class caveats are stated explicitly (Section 3.3.3), and no claim of clinical validity is made beyond what the data support.

The teamwork (PO(i)), communication (PO(j)), and management (PO(k)) outcomes are addressed through the execution of the project itself rather than through any single technical section. Finally, lifelong learning (PO(l)) is demonstrated by the literature review of Section 1.3, which spans 2015–2026 work across six research directions, and by the concrete future plan of Section 6.3.

5.2 Addressing Complex Engineering Problems

The thesis addresses a complex engineering problem because it involves research-based knowledge at the forefront of the discipline, wide-ranging technical issues, and non-obvious design decisions. Table 5.2 justifies each attribute of a complex engineering problem with evidence from this work.

Table 5.2. Complex Engineering Problem Solving through the Project & Thesis (EEE 4000)

Attribute	**Statement**	**How It Is Addressed**
Depth of knowledge required (P1)	Requires research-based knowledge, much of which is at, or informed by, the forefront of the professional discipline and that allows a fundamental-based, first principles analytical approach	Section 1.3: the solution draws on 2015–2026 state-of-the-art research (attention, SE, FiLM, TinyML); Section 2.5 derives the architectures from first principles.
Range of conflicting requirements (P2)	Involve wide-ranging or conflicting technical, engineering and other issues	The accuracy–footprint–latency trade-off of Section 3.3.11, and the conflict between permissive and record-disjoint evaluation (Section 3.4), are conflicting requirements resolved by measurement.

Table 5.2 continued from the previous page

Attribute	**Statement**	**How It Is Addressed**
Depth of analysis required (P3)	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models	Section 2.6: the fusion-beat treatment and the auxiliary feature vector are original formulations; Section 3.3.6 shows the naive solution ($A2$) fails and only a staged, analysed design works.
Familiarity of issues (P4)	Involve infrequently encountered issues	The S and F classes (Sections 1.2 and 3.3.6) are rare beats whose correct classification is unfamiliar to generic models; the Q-class single-patient caveat (Section 3.3.3) is an infrequently encountered statistical issue.

Table 5.2 continued from the previous page

Attribute	**Statement**	**How It Is Addressed**
Extent of applicable codes (P5)	Are outside problems encompassed by standards and codes of practice for professional engineering	The ANSI/AAMI EC57 standard (Section 2.8) governs reporting but does not prescribe an architecture or an imbalance strategy; Sections 3.3.4–3.3.7 show the standard must be combined with novel design work.
Extent of stakeholder involvement and conflicting requirements (P6)	Involve diverse groups of stakeholders with widely varying needs	Section 4.1: patients (early detection), cardiologists (alarm burden, trust in explanations), and low-income users (cost) have different needs; Section 3.3.5 quantifies the trade-off between detection and false alarms.

Table 5.2 continued from the previous page

Attribute	**Statement**	**How It Is Addressed**
Interdependence (P7)	Are high-level problems including many component parts or sub-problems	The pipeline integrates preprocessing, feature engineering, two architectures, imbalance handling, explainability, and deployment; the ablation of Section 3.3.7 shows the components are interdependent, with each block critical for a different class.

5.3 Addressing Complex Engineering Activities

The development of this system also constitutes a complex engineering activity. Table 5.3 justifies each attribute, drawing on Chapter 4 for the societal and environmental consequences (A4).

Table 5.3. Complex Engineering Activities Solving through the Project & Thesis (EEE 4000)

Attribute	**Statement**	**How It Is Addressed**
Range of resources (A1)	Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies)	Sections 2.3–2.7: three public databases, GPU training hardware, STM32 target hardware, Python/TensorFlow/TFLite software, and the two authors' and supervisor's expertise.
Level of interaction (A2)	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues	Sections 3.2.5 and 3.3.11: quantisation interacts with accuracy, the recurrent stage interacts with the evaluation protocol, and augmentation interacts with the fusion class; each interaction was resolved by experiment.

Table 5.3 continued from the previous page

Attribute	**Statement**	**How It Is Addressed**
Innovation (A3)	Involve creative use of engineering principles and research based knowledge in novel ways	Section 2.6: the staged fusion-beat pipeline and the MSFT-Net design (multi-scale depthwise-separable stem with SE and FiLM) are novel combinations of published principles, validated in Sections 3.3.6–3.3.7.
Consequences for society and the environment (A4)	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation	Chapter 4: the consequences for cardiac patients and the Bangladeshi healthcare system are significant; the risk of misdiagnosis and alarm fatigue is difficult to predict and is mitigated by statistical operating point analysis (Section 3.3.5).

Table 5.3 continued from the previous page

Attribute	**Statement**	**How It Is Addressed**
Familiarity (A5)	Can extend beyond previous experiences by applying principles-based approaches	The work extends standard CNN/LSTM practice to the record-disjoint protocol and to real microcontroller deployment, both of which go beyond the experiences reported in most of the surveyed literature (Section 1.3).

Chapter 6

Conclusion and Future Plan

This chapter closes the book. Section 6.1 restates the problem and the contribution of the work, Section 6.2 states honestly what the work does not cover, and Section 6.3 proposes specific next steps.

6.1 Conclusion

The problem addressed by this thesis was the unreliable and often inflated reporting of ECG arrhythmia classification performance, combined with the difficulty of running accurate classifiers on the small microcontrollers found in wearable monitors. To answer it, a hybrid CNN–BiLSTM–attention model and a lightweight multi-scale feature-augmented network (MSFT-Net) were designed, trained, and compared under a strictly record-disjoint, inter-patient partition of the MIT-BIH data, following the ANSI/AAMI EC57 class grouping.

Each objective of Section 1.5 was met. The first objective, designing the two architectures with rhythm, morphological, and template-similarity auxiliary features, was achieved in Chapter 2: MSFT-Net (167,090 parameters) attained 89.45% accuracy and 87.25% macro-F_1 under the inter-patient protocol, significantly outperforming the 3.4× larger recurrent hybrid (82.00% accuracy, 73.26% macro-F_1, $p \approx 0$ by McNemar test), while the same architecture family reached 98.59% accuracy under the permissive intra-patient six-category protocol. The second objective, honest evaluation under both protocols with statistical significance testing, was met in Chapter 3, and it produced the central finding of the thesis: the evaluation protocol changes the reported performance by roughly nine to ten points and even reverses the architectural conclusion, since the recurrent stage helps under the subject-mixed split but actively harms inter-patient generalisation. The third objective, component analysis and faithful explainability, was met by the twelve-arm ablation, which showed that the auxiliary feature vector is the engine of the model (removing it costs 0.2819 macro-F_1) and that the attribution maps, while plausibly localised to the QRS complex, fail the deletion-curve faithfulness test and must be treated as qualitative. The fourth objective, INT8 deployment on an STM32 microcontroller, was met with a measured per-beat inference latency of 185.057 ms, a 288.16 KB model occupying 56.28% of flash and 75.97% of RAM, and the full inter-patient accuracy preserved on-device.

The contribution of this work is threefold: (i) a protocol-correct, statistically disciplined evaluation of two competing ECG classification architectures under identical splits, quantifying for the first time in a single controlled study how much apparent performance is a by-product of the assessment regime; (ii) a compact, feature-augmented convolutional architecture (MSFT-Net) with a staged, evidence-based treatment of the resistant fusion class; and (iii) a demonstrated deployment path from deep learning model to real-time INT8 inference on actual microcontroller hardware, with measured rather than asserted deployability.

6.2 Limitations

The following limitations qualify the conclusions above.

First, the Q class rests on very few patients: its test support is a single held-out paced record, so its sensitivity and positive predictivity are de facto single-patient statistics whose real uncertainty is the between-patient spread, not the nominal binomial confidence intervals. Second, the CHF category of the intra-patient study comprises ECG segments derived from a database that reflects only hospitalised patients with congestive heart failure, recorded at a different sampling frequency than the arrhythmia database; the source-database probe confirmed a learnable database fingerprint in the integrated corpus, so the near-perfect separation of the CHF-derived category under the intra-patient protocol must not be interpreted as a cardiac-morphology finding or as evidence of a clinically generalisable CHF signature. Third, robustness was measured only with synthesised Gaussian noise and simulated baseline wander; this is not equal to validation on an independently collected ambulatory recording, so the level of real-world sensor and motion artefact to which the results generalise is untested. Fourth, the latency of the smallest MCU model (Table 3.14) was characterised on a general-purpose host, not on target embedded hardware, so its timing reflects the computational footprint rather than a demonstration of real-time on-device operation; the full-accuracy MSFT-Net was, however, profiled directly on the STM32. Fifth, both architectures were evaluated on data from a few public databases and a single record-disjoint partition; a completely out-of-hospital patient population would be a more rigorous test of generalisation than any within-database split can provide. Finally, the calibration and decision-threshold experiment showed that thresholds tuned on the validation partition do not transfer to the test partition, so any operating-point tuning performed for deployment must be verified on data from a more representative deployment population than an internal split can provide.

6.3 Future Plan

Several concrete next steps follow from the limitations above:

1. Extend the evaluation to an external, independently collected ambulatory cohort (for example a wearable single-lead database), so that the robustness and generalisation claims are validated on real sensor and motion artefacts rather than synthetic corruption.

2. Increase the Q-class support by adding more paced records from other public databases, so that the between-patient statistics of the Q class become meaningful rather than single-patient.
3. Improve the faithfulness of the attribution maps by training with explanation objectives (for example attention or saliency regularisation), and re-test with deletion curves and intersection-over-union against cardiologist-annotated intervals.
4. Investigate the hypothesis, raised by the reversal of the recurrent-stage benefit between protocols, that recurrent architectures trained on multi-subject data preferentially learn patient-identity-correlated temporal structure, for example by probing the learned recurrent representations for patient identity.
5. Add on-device probability calibration with abstention so the wearable can withhold low-confidence classifications instead of emitting unreliable decisions, and validate the chosen operating point on a deployment-representative population.
6. Extend the deployment work to continuous multi-hour operation on the STM32 (real-time acquisition, buffering, and communication stacks alongside inference), and measure battery life and power consumption in a wearable form factor.
7. Retrain the pipeline on multi-lead and 12-lead data and on additional arrhythmia classes beyond the five AAMI classes, to move the system towards broader clinical screening coverage.

These steps are the natural continuation of the work: each one follows directly from a limitation identified in Section 6.2 or from a finding of Chapter 3, and together they outline the path from the present prototype to a clinically validated, wearable cardiac monitoring system.

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Appendix A

Hyperparameter Tuning and Ablation Study Results

This appendix documents the model-selection experiments behind the final inter-patient configuration: the hyperparameter search space explored for MSFT-Net and the component-addition study that justified its two auxiliary components. The twelve-arm ablation of the hybrid model referenced in Section 3.3.7 is reported in Table 3.12 of Chapter 3 and is not repeated here. All results in this appendix are under the inter-patient protocol; the best configuration is reported in Chapter 3.

A.1 Search Grid

Table A.1 defines the search space from which 240 configurations were sampled during the random search; the best value found for each hyperparameter is listed.

A.2 MSFT-Net Component-Addition Study

The FiLM conditioning and RR-interval features of MSFT-Net were selected through a factorial component study. Starting from a baseline without either component, each configuration enables one or both, and every configuration was retrained and evaluated on the frozen inter-patient test partition. The resulting macro-F_1 values are reported in Table A.2; each component contributes a substantial gain on its own, and their combination is super-additive, reaching the 87.25% macro-F_1 of the final MSFT-Net (Tables 3.7 and 3.8). Per-class sensitivities and positive predictivities for this final model are given with their Wilson confidence intervals in Table 3.9.

Table A.1. Hyperparameter search grid for the inter-patient protocol.

Hyperparameter	**Values**	**Best**
Learning rate	$\{10^{-2},\ 10^{-3},\ 10^{-4}\}$	10^{-3}
Batch size	$\{32,\ 64,\ 128\}$	64
Stem filters	$\{16,\ 24,\ 32\}$	32
SE ratio	$\{4,\ 8,\ 16\}$	8
Dropout rate	$\{0.2,\ 0.3,\ 0.4\}$	0.3
FiLM conditioning	{on, off}	on
RR-interval features	{on, off}	on
Weight decay (L_2)	$\{0,\ 10^{-5},\ 10^{-4}\}$	10^{-5}

Table A.2. Effect of FiLM conditioning and RR-interval features in MSFT-Net (macro-F_1, inter-patient test partition)

Arm	**Configuration**	**Macro-F_1 (%)**
1	Baseline (no FiLM, no RR)	79.32
2	FiLM only	82.15
3	RR features only	83.89
4	Both (full MSFT-Net)	87.25

Appendix B

Dataset Statistics and Partition Details

This appendix provides the complete database-level statistics for all three PhysioNet databases used in this thesis, the exact record-disjoint partition lists, and the class distribution per partition.

B.1 Database Overview

Table B.1 summarises the source databases, their sampling rates, record counts, and total duration.

B.2 Record-Disjoint Partition Lists

The de Chazal-style DS1/DS2 partition is applied only to the MIT-BIH records. NSR and CHF records are split by record with no overlap between partitions.

B.2.1 MIT-BIH Arrhythmia Database

Records 102, 104, and 107 are paced records retained in DS1 for training regularisation only; record 217 is the single held-out paced record in DS2 and supplies the entire Q-class test support of Table 3.6. Consistent with Section 3.3.1, paced beats contribute to the Q class only and are never scored as N/S/V/F.

B.2.2 MIT-BIH Normal Sinus Rhythm Database

B.2.3 BIDMC Congestive Heart Failure Database

B.3 Class Distribution per Partition

The beat-level class distributions for each partition are reported in the main body and are not repeated here: Table 3.6 gives the five-class inter-patient composition (train/validation/test), and Table 3.2 gives the six-category intra-patient corpus from raw extraction through capping to the SMOTE-balanced training set. The corpus is severely imbalanced – before quota sampling and balancing, normal beats outnumber fusion beats by roughly 19:1 (15,306 vs. 802 beats) – which motivated the three-way comparison of imbalance strategies in Section 2.7. As the ablation of Section 3.3.7 shows, training-only minority augmentation with

Table B.1. Source database statistics.

Database	Records	Fs (Hz)	Total Beats	Duration
MIT-BIH Arrhythmia	48	360	109,494	30.5 h
MIT-BIH NSR	18	128	18,266	42.8 h
BIDMC CHF	15	250	15,330	36.0 h
Total	**81**	—	**143,090**	**109.3 h**

Table B.2. MIT-BIH record-disjoint partition (de Chazal).

Partition	Record Numbers
DS1 – Train	101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 209, 102, 208, 104, 107
DS1 – Val	215, 220, 223, 230
DS2 – Test	100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234, 217

Table B.3. NSR record partition.

Partition	Record Numbers
Train	nsr001–nsr012 (12 records)
Val	nsr013–nsr015 (3 records)
Test	nsr016–nsr018 (3 records)

Table B.4. CHF record partition.

Partition	Record Numbers
Train	chf001–chf009 (9 records)
Val	chf010–chf012 (3 records)
Test	chf013–chf015 (3 records)

no global re-weighting was found to be the best strategy under the record-disjoint protocol; class weighting and SMOTE were tested and rejected.